



FAQs on Valuation

Shashank Gupta



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Frequently Asked Questions on Valuation

1 General

1.1 Why is there a need for the market to correct its mistakes?

By investing money in anything, we are foregoing purchasing power today to get higher purchasing power in the future. So, the only reason to invest money today is to get money back which would allow us to buy more goods in the future. There are only two ways in which this can be achieved from an investment:

- Capital return (through dividends and buy-backs); and
- Capital appreciation.

Suppose the market is undervaluing a firm which is generating immense amounts of cash flow. If the market does not realise its mistake, one way for an investor to get his money back would be for the firm to declare a huge special dividend. But, as a passive investor, the investor may not be able to affect the dividend policy of the firm. Therefore, he is highly reliant on the capital markets to correct their mistakes.

Also, in today's world, only a fraction of the total return from stocks comprises dividends. Most return comes from capital appreciation, which obviously is dependent on the market correcting its mistakes.

And then, you also think hard about opportunity costs. To sit on a stock for 10 years without it doing much is easier said than done, let alone for 20-30 years. And in any case, if the markets are not doing what they are supposed to do as per the model for such a long time, it is highly probable that the mistake is with the model and not the market.

1.2 How do you decide whether to use certainty equivalent cash flows which can be discounted at the risk-free rate vs. risk adjusted discount rate?

The debate between using certainty equivalent cash flows vs. risk adjusted discount rate arises from the fact that Warren Buffett uses certainty equivalent cash flows and discounts them at the risk-free rate of the economy. The reason cited for the above is that it eliminates the need to estimate a discount rate.

However, this approach is becoming harder to follow today because of the following reasons:

- Gradual shift in the composition of the cash flows of companies. If you look at the concept of a financial balance sheet, a larger proportion of cash flows are now reliant on growth assets than existing assets.
- Secondly, increase in macro-economic risks since the global financial crisis has also made it more difficult to assess certainty equivalent cash flows for a company.

1.3 The role of markets

<https://www.youtube.com/watch?v=5V79HrOVri8>

2 DCF: Risk free rates

2.1 What is a risk-free return?

A risk-free return is one where the actual return is equal to the expected return, i.e. there is no variance around expected return.

For an investment to be risk free, it has to satisfy the following two conditions:

- It should not have any default risk; and
- It should not carry any reinvestment risk.

The following three elements form the foundation for thinking about risk-free rates:

- **Time horizon:** Risk free rates depend on the timing of the cash flows. For instance, for a 3-month cash flow, the correct risk-free rate would be the 91-day T-Bill.
- **Currency:** Risk free rates are currency specific.
- **Not all governments are risk-free:** Some governments carry default risk and thus, the local currency bond rate cannot be considered as the risk-free rate for the country.

2.2 Why do we use a 10-year T-Bond rate to calculate the risk-free rate?

One of the conditions of a risk-free rate is that the investment should not carry any reinvestment risk. Thus, we need to match the duration of the riskless asset to the duration of the cash flows that we discount. For instance, if we are assessing a cash flow 3 months from now, the correct risk-free rate would be the yield on a 91-day T-Bill.

When we value the equity in a company on a going concern basis, we are estimating cash flows for long periods of time (sometimes into infinity) and consequently, we should use the longest government bond rates available. In many instances, this would be the 30-year government bond rate.

However, we generally end up restricting ourselves to the 10-year government bond rate to compute the risk-free rate. The reason we do this is because using the 10-year rate makes our life a lot easier as we move along in the valuation process. If the end game was to only come up with the cost of equity, we would not hesitate to use the 30-year rate as the risk-free rate. But, a cost of equity is only one of the components of the valuation process. As we move along, we also need to compute a cost of debt. And the cost of debt is derived as follows:

$$\text{Cost of Debt} = R_f + \text{Default Spread on Corporate Bonds}$$

As a matter of consistency, **the duration of the corporate bond must match the duration of the risk-free asset.** The way we compute the default spread on a corporate bond for a particular rating is to look at other corporate bonds with the same rating and comparing them with the risk-free rate. This is where the 30-year bond breaks down, since there are not many 30-year corporate bonds. Hence, we end up using the 10-year bond as a proxy for long term cash flows.

2.3 What are the steps in calculating the risk-free rate for a particular currency?

The risk-free rate for a particular currency can be written as:

$$\text{Risk free rate} = 10 \text{ year local currency bond rate} - \text{Default spread}$$

For a country whose local-currency bonds are rated 'AAA', we assume the default spread to be 0 and accordingly, the 10-year local currency bond becomes the risk-free rate for such a country.

However, for countries that are not rated 'AAA', we need to compute a default spread and reduce it from the local currency bond rate to arrive at the risk-free rate. The ways to compute the default spreads is discussed in the next question.

2.4 How to compute the default spreads in order to compute the risk-free rate for a particular currency?

There are 3 ways to compute the default spreads:

Method 1: Check whether the country has sovereign US\$ denominated 10-year bonds. If it does, the default spread is computed as:

$$\text{Default spread} = \text{Local bond rate (in US\$)} - \text{US 10 year T bond rate}$$

Method 2: Check whether the country has a sovereign CDS spread. If it does, the default spread is computed as:

$$\text{Default spread} = \text{CDS spread for country in question} - \text{US CDS spread}$$

Method 3: This method is used when both the previous methods fail. In this method, we compute the default spreads (using either Method 1 or 2) for all countries which have the same rating as the rating of the concerned country. The average of these spreads can be used as the default spread for the concerned country. For instance, if the sovereign rating for your country's local currency bonds is Baa3, the default spread will be the average default spread of all countries whose sovereign bonds are also rated Baa3 and which also have a US\$ denominated sovereign bond/ CDS.

2.5 How to compute the default spreads for countries which do not have a rating?

In such a scenario, we can impute the default spread using PRS (Political Risk Scores). We compare the concerned country's PRS score with other countries which have a similar PRS and assign it a default spread based on the average of default spreads of those countries.

2.6 Why are risk-free rates different in different currencies?

The difference in risk-free rates arises because of differences in inflation across countries. One thing to keep in mind is that the risk-free rate and the cash flows should be of the same currency in a valuation.

Also, if we know the risk-free rate in Country A (R_{fA}) and the expected inflation in Country A (EI_A) and Country B (EI_B), we can calculate the risk-free rate for Country B (R_{fB}) as follows:

$$R_{fB} = [1 + R_{fA}] * \frac{[1 + EI_B]}{[1 + EI_A]} - 1$$

One of the most important things to keep in mind regarding risk free rates is that they are specific to the currency in which you are trying to calculate them.

Also, by definition, for an investment to be risk free, it has to satisfy the following two conditions:

- It should not have any DEFAULT risk; and
- It should not have any REINVESTMENT risk.

For the purpose of your question, we are concerned with the first of those conditions, i.e. that a risk free rate is one which does not carry any default risk.

Once you take out the default risk from the equation, it follows that the risk free rates in various currencies should be such that an investor would be indifferent between investing in one currency or the other. That is, it would make no economic difference if an investor is invested in a riskless investment in Indian Rupees or whether he is invested in a risk-less investment in US dollars (assuming such investments exist). i.e. they would have the same amount of dollars after a year's investment by investing in a riskless Indian Rupee investment had they investment the same amount in a riskless US dollar investment.

It therefore follows that, for that condition to exist, the only difference in risk free rates across currencies has to be on account of differing inflation between currencies. In our example, with the given inflation rates, if you were to invest in an Indian Rupee investment yielding 9.655%, you would be in the same position had you investment in the riskless US dollar investment.

2.7 The concept of real risk-free rates and some other issues?

Real risk-free rates should be used only when discounting real cash flows. Generally, a real risk-free rates and real cash flows are used when the inflation is abnormally high in the currency in which you wish to do a valuation. TIPS rate is an example of a real risk-free rate.

A security will provide a real risk-free return if:

- it does not carry default risk; and
- guarantees a real return.

Can the TIPS rate be used as the real-risk free rate across countries?

The TIPS rate can be used as a real-risk free rate across countries as long as we assume that capital flows freely. This is because a country offering a higher 'real' return will attract capital pushing those rates down towards equilibrium. For instance, let's assume that the real return in US is 2% while it is 5% in India. If you are an investor in US assets, you will likely take money out of the US and start investing in India instead. When many investors start doing this, yields on Indian assets will start to go down and those on US assets will start rising, with the consequence that an equilibrium will be reached with no country offering an "extra" real return. Hence, it is possible to use the TIPS rate as the real risk-free rate across countries.

However, the assumption about free capital flows may not be appropriate in case of some countries. An alternative approach for such a country would be to set the real riskless rate as the '**Expected real growth rate in long term**'.

2.8 What is a synthetic risk-free rate?

A synthetic risk-free rate is what a risk-free rate should be given the growth and inflation expectations of an economy. It can be written as:

$$\text{Synthetic risk free rate} = \text{Real growth rate} - \text{Expected inflation}$$

We can use this rate even if we are uncertain about the expected inflation since the same risk-free rate is used in maturity to project cash flows. Accordingly, inflation errors will largely cancel out in a valuation.

2.9 Should you normalise risk-free rates?

No, risk free rates should never be normalized. The current risk-free rate acts as an actual opportunity cost of investing capital in riskier securities, TODAY. Thus, if you find equities as a class to be overvalued based on the choice of a risk-free rate then, you forgo investing in equities and invest your funds AT THE RISK-FREE RATE. However, if you cannot invest at a 'normalised' risk-free rate.

For example, suppose the current risk-free rate is 2% but the average risk-free rate over the previous 5 years is 4%. If an investor uses 4% as their risk-free rate, they might find equities to be overvalued (a higher discount rate leads to lower value, other things remaining constant). If they then decide to not invest in equities, they will not be able to invest risk-free at 4% because risk-free rates at that time are 2%.

3 DCF: Equity Risk Premiums

3.1 What is an equity risk premium?

An equity risk premium is the **price of risk**. It represents that rate which needs to be offered **over and above the risk-free rate** to lure investors to invest in equities as a class.

An equity risk premium is a receptacle (bucket) of every hope and every fear regarding equities in the markets at a particular point in time.

It is a function of:

- How risk averse investors are collectively; and
- How much risk they see in the average equity.

3.2 What are the ways of measuring an equity risk premium?

There are two ways in which an equity risk premium can be measured:

- Historical risk premium; and
- Implied equity risk premium.

Historical risk premium: It is the premium that stocks have historically earned over riskless securities. It is sensitive to:

- How far back in history should we go?
- Whether to use T-Bills or T-Bonds?
- Whether to use geometric or arithmetic averages?

Implied equity risk premium: Implied equity risk premium looks at what people are actually paying for bearing the risk of investing in equities. Although it is volatile, it is a forward-looking number and dynamic.

3.3 What are the potential problems with using a historical risk premium?

A historical risk premium suffers from the following problems:

- It is highly **subjective** in that the person computing must make choices.
- It is **backward looking**.
- **High standard errors** resulting in wide ranges for the premiums to be statistically significant.
- Estimates move **counter intuitively**, i.e., down after a crisis and up after prosperity.
- Suffers from **survivorship bias** when computed based only on US data (because US was the most successful market of the 20th century. Using it means you are assuming that the market for which you are computing the risk premium is going to be the best performing market going forward); and
- Assumes **mean reversion**. This may or may not be true.

3.4 What is an implied equity risk premium and how is it computed?

An implied equity risk premium is a forward looking premium that is derived by looking at what market participants are actually demanding given the following inputs:

- The level of the index;
- The cash flows;
- The growth in those cash flows; and
- The risk-free rate of return.

Computation of implied equity risk premiums:

The calculation of implied equity risk premiums uses the framework of discounting cash flows on a coupon bond to come up with the yield of that bond where:

- The current market price is set equal to the current level of the index (e.g. S&P 500).
- The earnings on the index play the role of the coupons (only difference being that earnings can be volatile unlike coupons on a bond). These can be considered as any of the following:
 - The actual dividend + buyback yield of the index;
 - The earnings yield of the index; or
 - Yield calculated on the basis of free cash flow to equity of companies comprising the index.
- Since the earnings are not guaranteed, we need to estimate a growth rate in these earnings. Note that in the calculation of implied ERP, we use consensus growth rates instead of fundamental growth rates.
- Once we have the above inputs, we can calculate the rate of return which equates the present value of the cash flows to the current level of the index (with the help of the goal seek function in excel).
- Lastly, we need to deduct the risk-free rate from this number to come up with the estimate of the implied equity risk premium.

Note that the process described above is only sufficient for calculating the equity risk premium for a market which does not have any default risk (i.e., AAA-rated countries also called mature markets). For countries that carry default risk, we need to add a country risk premium (discussed later).

3.5 Why do we use consensus growth rate (instead of fundamental growth rate) to compute implied ERP?

Implied ERP is one input in valuation where we look to what the market "is actually" doing rather than what it "should" be doing.

Another way to think about the Implied ERP is that it is a **measure of the risk averseness or otherwise of the market participants** at a given point in time as opposed to an investor's own risk averseness. And we can find that out only if the inputs going into the Implied ERP calculation come from the market itself.

Basically, what matters in an Implied ERP is **what others think** about the equity markets collectively and not what you as an individual think about it. That is why if you see, all other

inputs in an Implied ERP calculation also come from the market - the level of the index, the current dividend yield, the risk-free rate, etc.

3.6 What is the relation between the implied equity risk premium and its various inputs?

The impact of the movement in an input (keeping other inputs constant) on the equity risk premium is as follows:

Increase in Variable (ceteris paribus)	Impact on Implied Equity Risk Premium
Value of the index	Decreases
Cash flows	Increases
Risk free rate	Decreases
Growth rate	Increases

3.7 How to compute a country risk premium for emerging market companies, i.e. companies which have default risk?

To compute the full equity risk premium for countries which have an element of default risk, a measure of country risk needs to be added to the mature equity market premium ("Mature ERP"). Thus, for an emerging market country:

$$\text{Equity Risk Premium} = \text{Mature Market ERP} + \text{Country Risk Premium (CRP)}$$

There are four ways in which country risk can be incorporated into the above analysis.

- **Ex-McKinsey Approach – No CRP:** In this approach, CRP is set equal to 0 on the argument that country risk can be diversified away (by investing in different countries) and therefore, an investor should not be compensated for bearing this risk.

However, this is an empirical question and for this to be true, the correlation amongst equity indexes across the world needs to be low. Today, this is not the case.

- **Default spread as CRP:** In this approach, the CRP is set **equal to the default spread** computed for the country at the time of estimating its risk-free rate. Thus,

$$\text{CRP} = \text{Default Spread}$$

However, this approach assumes that the debt issued by the government of the country is **as risky as** the average equity of that country.

- **Goldman approach – relative equity volatility approach:** This approach calculates the total risk premium for the emerging market by **scaling it based on the relative volatilities** of the equity markets of the emerging market in question and that of a mature market. The equity risk premium for a mature market is then deducted from this number to arrive at the CRP. Thus,

$$\text{Total Risk Premium} = \text{Mature Market ERP} * \frac{\text{SD in Equity of Emerging Market}}{\text{SD in Equity of Mature Market}}$$

$$\text{CRP} = \text{Total Risk Premium} - \text{Mature Market ERP}$$

However, this approach assumes that the liquidity in the equity of the emerging market is similar to that of a mature market. In case this is not true, the standard deviation in the emerging market equity may turn out to be lower than that of a mature market thereby resulting in a total equity risk premium lower than that of a mature market (which should not be the case).

- **Melded approach:** In this approach, the default spread is scaled based on the relative volatilities of the equity market of the emerging market and the debt markets of that same country. Thus,

$$\text{CRP} = \text{Default Spread} * \frac{\text{SD in Equity of Emerging Market}}{\text{SD in Debt of Emerging Market}}$$

This approach assumes similar liquidity in the equity and debt markets of the country. You need to check if this assumption holds for the country for which you are computing the CRP.

3.8 How to measure the exposure to country risk premium, i.e., from Country risk premium to corporate risk premiums?

Once we have the CRP, we need to measure what the risk premium would be for the company which is being valued. There are 3 ways in which we can measure the exposure of a company to country risk:

- **Approach 1:** Assume that every company in the country is equally exposed to country risk. In this approach, the cost of equity will be computed as:

$$\text{Cost of Equity} = R_f + \beta * \text{Mature ERP} + \text{CRP}$$

- **Approach 2:** Assume that a company's exposure to country risk is similar to its exposure to other market risk. In this approach, the cost of equity will be computed as:

$$\text{Cost of Equity} = R_f + \beta * (\text{Mature ERP} + \text{CRP})$$

- **Approach 3:** Treat country risk as a separate risk factor and allow firms to have different exposure to country risk. In this approach, the cost of equity will be computed as:

$$\text{Cost of Equity} = R_f + \beta * \text{Mature ERP} + \lambda * \text{CRP}$$

Some important points to note about country risk are:

- **Source of revenues:** Other things remaining the same, a company should be more exposed to risk in a country if it generates more of its revenues from that country.
- **Manufacturing facilities:** Other things remaining the same, a firm that has all its production facilities in a "risky country" should be more exposed to country risk. The problem will be accentuated for companies that cannot move their production facilities (mining and petroleum companies, for instance)
- **Use of risk management products:** Companies can use both options/ futures markets and insurance to hedge some or a significant portion of country risk.

- **Government "national" interests:** There are sectors that are viewed as vital to the national interests, and governments often play a key role in these companies, either officially or unofficially. These sectors are more prone to country risk.

3.9 What are some of the ways of computing separate exposure to country risk (i.e. λ)?

There are a couple of ways in which you can measure exposure to country risk:

- One way is to run a regression of stock returns for the company against the CDS spread/ Sovereign Bond of the country in which the company is incorporated.
- Another way is to measure how much an average company generates revenues in a country and compare your company's revenues in that country.

3.10 Should an investor be compensated for country risk? In other words, can country risk be diversified away?

The argument that there should be no compensation for country risk because it can be diversified was used by McKinsey in its valuation text. However, this argument has now been dropped.

Whether investors can diversify country risk is a matter of evidence. This argument may have worked in the past as economies were not correlated with one another. Globalisation, however, has changed this over the last couple of decades. As companies go global, correlations among countries have also started to rise with the consequence that it is much more difficult for investors to "diversify" country risk. Prof. Damodaran has this to say about the issue – Mumbai is no more immune to what happens in Sao Paulo.

Accordingly, a component of country risk needs to be incorporated in the ERP.

3.11 How can equity risk premiums help in deciding asset allocation?

One possible application of implied equity risk premiums is in deciding the asset allocation – i.e., whether the equity risk premiums at a particular point in time are high enough for investors to invest in equities as a class.

One can use the following measures to check whether the equity risk premium is commensurate or not:

- Compare the implied equity risk premium to the default spreads of investment grade bonds. Obviously, the bonds should have a lower spread than the equity risk premium but there are times when the two have converged (during the dot-com boom), giving an indication that one should be cautious.
- Another way is to compare equity risk premiums across markets if you are a global investor. For instance, you could compute an equity risk premium for the Indian market (based on relevant data from the Indian market) and see how it compares to the premium that you get by adding Indian country risk to a mature market premium. A big divergence between the two may indicate that Indian investors are not adequately providing for the country risk.

3.12 Is the US risk premium overestimated since the ERP is calculated from the S&P 500 in which companies get revenues from countries other than the us as well?

There is a possibility that the 'pure' US risk premium is being overestimated since companies in the S&P 500 draw about half of their revenues from outside the US. However, currently most of those revenues come from Canada and Developed European countries (which are rated 'AAA').

As exposure of US companies shifts increasingly to emerging markets, one would have to think of ways to adjust for this as exposure.

3.13 Why cash flows to investors in the dynamic model for ERP only include stock buybacks and dividends? Should it not also include the rest of the earnings that is kept in the business?

The aspect of why we consider only dividends and stock buybacks as "cash" available to shareholders and not the remaining earnings would become clear as we get into the Cash flows and Growth section of the course. However, let me try and answer this for you.

The earnings of the S&P 500 is nothing but the consolidation of the earnings of the companies in the S&P 500 weighted by market cap. The individual company earnings are BEFORE any capital requirements and AFTER depreciation.

Now, if a company has to grow, generally it would have to invest something over and above the depreciation expense in the income statement, otherwise, it'll keep growing while the invested capital base of the company shrinks, a near impossibility.

So, in the forward looking ERP, what Prof. Damodaran does is that he recognises that the actual cash flowing out of the companies to the investors right now is the dividend + stock buybacks, but over time he adjusts the payout ratio to determine a fundamental level of earnings that can be passed onto the shareholders given assumptions around ROC, long term growth and reinvestment requirements.

To your point, the forward-looking ERP model does take into account the earnings kept within the company through a HIGHER growth rate of earnings, which then get reflected in my terminal value.

3.14 Whether dividends and buybacks are true representation of FCFE?

If I understand you correctly, your hypothesis is that when valuing the market (S&P500)/ computing the market's ERP, current dividends and buybacks may not be reflective of "true" FCFE since they may have to be diverted to pay off debt "in the future".

Well, that would be true if you feel that the market as a whole is over levered and therefore, companies on the whole would need to "pay down debt" to reduce leverage on a net basis over time. I don't have the data for this, but my hunch would be that at an aggregate level, under leverage and over leverage would even out even though there would be significant differences at company-level debt ratios.

When looking at an individual company, however, I agree that current debt ratios may not be reflective of "ideal". That is why, when working with individual companies, it is better to work

with FCFF than FCFE – the big assumption when you work with FCFE is that the debt ratio remains constant. Working with FCFF (where you can) takes care of this issue and allows you to change debt ratios in the future to a level what you feel is right.

The usual metric that gets a lot of attention (both in the media as well as in investing circles) is the Corporate Debt to GDP which is probably at highest levels. However, there is a fundamental error with that metric in the sense that one is a flow variable (GDP) and one is a stock variable (Corporate Debt). And usually, such metrics break down in their "predictive" power over time.

I guess a more relevant metric to look at might be the overall debt/ equity across time and see whether there are significant differences in the current ratio vs. history. Again, I do not have this data handy (I probably should have), but I will look for it and share if I can find. As I said, these are all very valid points and I don't disagree that if debt ratios are significantly out of line with history, either up or down, then there is a case of adjusting the current dividends and buybacks to reflect what is likely to happen in the future. At the end of the day, the market as a whole is not any different from an individual company and the logic, we use to value individual companies should/ can be applied to valuing the market as a whole.

3.15 Can the difference between earnings yield and the risk-free rate be used as the equity risk premium?

The difference between the earnings yield and the risk-free rate as a measure of the equity risk premium is a short cut and can be a good proxy. However, this measure misses growth. Thus, it is likely to underestimate the ERP, especially in a year like 2020 where earnings are depressed, and earnings growth are expected to be high in the future.

4 DCF: Beta

4.1 What is a Beta (β) and what does it measure?

Beta measures how risky an asset is relative to the average risk investment. It measures only macro-economic risk exposure of a company and not firm specific risk (CAPM assumes that you have a diversified portfolio).

4.2 How can beta be estimated?

There are two ways of estimating β :

- **Regression beta:** Calculated by regressing the returns on a stock against some market index:

$$R_j = a + b * R_m$$

- **Bottom-up beta:** This beta is computed using the fundamental characteristics of the firm (discussed in detail later).

4.3 What are the drawbacks of a regression beta?

A regression beta suffers from the following three problems:

- It has a **high Standard Error**, making it meaningless;
- It reflects the firm's business mix over the period of the regression and **not the current mix**; and
- It reflects the firm's average financial leverage over the period of the regression than the **current leverage**.

4.4 What are bottom-up betas?

Since betas are a measure of risk, it can be argued that betas do not come from regressions but **from choices you make as a company**. There can be 3 sources of risk in a firm:

Nature of product or service offered	Operating leverage	Financial Leverage
Other things remaining the same, the more discretionary the product or service, the higher the beta.	Other things remaining the same, the greater the proportion of costs that are fixed, higher the beta.	Other things remaining the same, the greater the proportion of capital that a firm raises from debt, the higher its equity beta will be.
• Cyclical companies should have higher betas than non-cyclical companies. • Similarly, luxury goods manufacturers should have higher betas than	• Firms with high infrastructure needs and rigid cost structures should have higher betas than firms with flexible cost structures.	• Highly leveraged firms should have higher betas than firms with less debt.

basic goods manufacturers. • High priced goods/services firms should have higher betas than low priced goods/services firms. • Growth firms should have higher betas.	• Smaller firms should have higher betas than larger firms. • Young firms should have higher betas than more mature firms.	
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4.5 What are the steps in computing a bottom-up beta?

A bottom-up or fundamental beta for a company can be computed using the following steps:

- **Step 1: Start with the Beta of the business that the firm is in**

Suppose your company has two businesses: Steel and Chemicals. Find as many publicly traded Steel companies and chemical companies as possible. Here one may have to apply a filter¹ to see if those companies are involved in other businesses as well.

Find out the regression betas for these companies and take a **simple average**² of these betas. This beta is a levered beta since the regression is run based on the stock prices against the index. *Since leverage would have an impact on the earnings of the company and thereby its stock price, we can say that leverage has already been factored in this beta.*

Why not first calculate the unlevered beta for individual companies and then average the unlevered betas?

When you have a noisy number, doing mathematical operation on that noisy number will make it noisier. An individual regression beta can be all over the place and adjusting that number is going to exaggerate the effects of those operations.

Average first and then apply adjustments on the average to find the unlevered beta.

- **Step 2: Adjust for Operating Leverage**

If we have the data, we can adjust for operating leverage. However, most of the times we are not likely to have the information regarding the fixed and variable costs of the businesses.

- **Step 3: Calculate the average Debt-Equity Ratio**

Since we need to adjust for financial leverage, we need to compute the average debt-equity ratio of companies that were used in computing the levered beta in Step (a) above.

- **Step 4: Calculate the Unlevered Beta for the Company (pure business beta)**

The first step in adjusting for financial leverage is to unlever the beta with the average debt-equity ratio of the companies used in the sample. This is done as follows:

¹ Such as categorising companies as steel/ chemicals if more than 75% of their revenue comes from that business.

² Using weighted averages would undercut the benefits of averaging, since it counts larger firms more and smaller firms less.

$$\text{Company Unlevered Beta } (\beta_{UC}) = \frac{\beta_L}{[1 + (1 - \text{tax rate}) * \text{Average DE Ratio}]}$$

- **Step 5: Clean-up for cash to calculate the pure business beta**

Step (d) will give you the unlevered beta for the company and not necessarily for the pure business (since companies have a cash balance as well and the beta of cash is 0). To calculate the pure business beta here is what we need to do:

$$\text{Business Unlevered Beta } (\beta_{UB}) = \frac{\beta_{UC}}{\left(1 - \frac{\text{Cash}}{\text{Firm Value}}\right)}$$

- **Step 6: Calculate Levered Beta for the company**

In this step we calculate the levered beta to reflect the debt-equity ratio of the firm for which the beta is being computed. There are two ways in which we can do this:

Conventional Approach	Debt-Adjusted Approach ³
Assumes that debt carries no market risk (i.e., it has a beta of 0 ⁴).	Assumes that some of the risk is borne by debt but requires an estimation of beta of debt (which is extremely difficult).
Levered Beta (β_L) = $\beta_{UB} * [1 + (1-t) * (\text{debt-equity ratio})]$	$(\beta_L) =$ $\beta_{UB} * [1 + (1-t) * (\text{debt-equity ratio})] - B_{DEBT} * (1-t) * (\text{debt-equity ratio})$

4.6 What are benefits of using a bottom-up beta?

Listed below are some of the benefits of using a bottom-up/ fundamental beta:

- **Low standard error:** Even though we use regression betas as a starting point for calculating bottom-up betas, the standard error will be significantly lower than the standard error in a single regression beta. Roughly speaking, the standard error of a bottom-up beta can be estimated as follows:

$$\text{Standard error of bottom up } \beta = \frac{\text{Average standard error across individual } \beta}{\sqrt{\text{No. of firms in the sample}}}$$

- **Changes in business mix and financial leverage:** It can be adjusted to reflect changes in business mix and financial leverage of a firm whereas regression betas reflect the past.
- **Market data not required:** Bottom-up betas can be estimated even when historical stock prices are not available for the company being valued. Therefore, it allows us to value IPOs, private businesses, etc.

³ Prof. Damodaran estimates the beta of debt for highly risk companies in his corporate finance spreadsheets.

⁴ It doesn't mean that the debt is riskless but that it bears no market risk. And this calculation breaks down in riskier companies.

4.7 Can betas be negative?

Yes, if an investment is exposed to a macroeconomic risk such that it does well if that risk unfolds (while the rest of the market does badly). Betas can be negative, but it is rare to have a negative beta stock (e.g. gold mining companies).

If you get a beta that is negative for a business, then it is highly unusual and it is likely to be some kind of hedge against some macroeconomic risk.

4.8 Is it ok to have a negative beta stock in your portfolio?

One of the consequences of having a negative beta is that the investment's expected return falls below the risk-free rate. A question then arises as to why would anyone settle for anything less than the risk-free rate? The reason investors would buy such a stock is because it is like buying insurance (when inflation goes up an investor hopes that his/ her gold mining stock goes up and protects the rest of the portfolio).

Having a negative beta stock in a portfolio is OK as long as an investor is diversified. If an investor is not diversified and has all of his investment in a negative beta investment (such as gold), then it is like buying fire insurance on a property when he doesn't own any property. When an investor has nothing to insure, he will keep paying the premiums, but the payoff will never come.

4.9 Can betas be compared globally?

Yes, because betas are measures of relative risk. So, the average beta across all US companies is 1 and average beta across all African companies is also 1. You can't compare PE ratios/ interest rates across countries, but you can compare betas.

4.10 The concept of marginal investor and why only consider market risk?

How can a marginal investor compel managers to do anything? The board of directors can but replacing a board is not that easy. This is all you need. The marginal investors will sell their shares in the company if managers are too conservative (and consider total risk in the company when evaluating projects). That will push the stock price down. To the extent that managers care about the stock price, they will mend their ways and start looking at market risk. If they don't, you have to hope that activist investors or acquirers show up and try to change management.

5 DCF: Cost of debt

5.1 What is the definition of debt?

For an item to be classified as debt, it has to meet three criteria:

- Give rise to a contractual commitment, that has to be met in good times or bad;
- The commitment is usually tax deductible; and
- Failure to honour that commitment can cost you control over the business.

Using this criterion, all interest-bearing commitments, short term and long term, are clearly debt (**including leases**). The following items may also be considered as debt, if they meet other criteria:

- **Accounts payable and supplier credit:** Only if you are willing to make the implicit interest expenses (i.e. the discounts lost by using credit) explicit;
- **Under-funded pension and health care obligations:** Only if there is a legal requirement that you cover the underfunding with fixed payments in future years.

5.2 What is the meaning of cost of debt and how is it measured?

The cost of debt is the rate at which you can borrow:

- long term +
- today

It does not matter what duration of debt the company currently has or the rate at which it had borrowed money earlier. That is, **book interest should not be used**. It reflects:

- Your default risk; and
- Level of interest rates in the market.

Cost of debt for a firm in developed market is as follows:

$$\text{Pretax cost of debt} = R_f + \text{Corporate default spread}$$

Companies that are **incorporated and based in emerging markets** are **exposed to country risk** in addition to the corporate default spreads. Accordingly, we need to estimate what **portion of country risk needs** to be incorporated into the default spread:

$$\text{Pretax cost of debt} = R_f + \text{Corporate default spread} + \text{Portion of CRP}$$

5.3 How is a cost of debt computed?

There are three ways in which a cost of debt can be computed for a firm:

- **Yield-to-maturity on a bond issued by the firm:** The problem with this approach is that there are very few firms that have long term straight bonds that are liquid and widely traded.
- **Look up the bond rating for the firm and estimate a default spread:** A potential with this approach is that some firms may not be rated.

- **Estimate a synthetic rating and compute a default spread based on that rating:** This approach can be used even if the firm does not have a bond rating.

5.4 How to compute a synthetic rating for a firm?

A synthetic rating for a firm can be estimated using the financial characteristics of the firm. A simple ratio to use is the interest-coverage ratio:

$$\text{Interest coverage ratio} = \frac{EBIT}{\text{Interest expense}}$$

The interest coverage ratio can then be converted into a rating based on a look-up table⁵ as below:

Spread Table			
If Interest Coverage Ratio is			
>	<= to	Rating is	Spread is
12.50	100000	Aaa/AAA	0.60%
9.50	12.499999	Aa2/AA	0.80%
7.50	9.499999	A1/A+	1.00%
6.00	7.499999	A2/A	1.10%
4.50	5.999999	A3/A-	1.25%
4.00	4.499999	Baa2/BBB	1.60%
3.50	3.999999	Ba1/BB+	2.50%
3.00	3.499999	Ba2/BB	3.00%
2.50	2.999999	B1/B+	3.75%
2.00	2.499999	B2/B	4.50%
1.50	1.999999	B3/B-	5.50%
1.25	1.499999	Caa/CCC	6.50%
0.80	1.249999	Ca2/CC	8.00%
0.50	0.799999	C2/C	10.50%
-100,000.00	0.499999	D2/D	14.00%

However, the above look-up table is based on default spreads of US companies (i.e. it is a dollar-based spread). And it works well only as long as the **inflation profile of your country is similar to the US**. In order to adapt the above table to higher inflation countries, **we can scale up the default spread** with the ratio of risk-free rate for that country to the US risk-free rate.

$$\text{Corporate default spread} = \text{Spread in lookup table} * \frac{R_f \text{ for Country A}}{US R_f}$$

5.5 Why do we add country default spread (apart from corporate default spreads) to come up with the cost of debt for a company?

The reason we add the country default spread (or at least some portion of it) is because practically when the company goes out to borrow money, it will carry some of the baggage of the country it belongs to or does business in.

⁵ Available on Prof. Damodaran's website; This data is as of 1 January 2017 for US companies.

Think in terms of a company which operates in an emerging market in turmoil (say Greece in 2013). No matter how great the company's financials are, it would be impossible for it to borrow at the same rate (in USD terms) than an otherwise similar company in a developed market.

5.6 Which tax rate (effective or marginal) should be used to compute the after-tax cost of debt?

For the purpose of computing cost of capital, we consider after-tax cost of debt since interest expenses on debt are tax deductible. We should always use the marginal rate of tax since **interest saves taxes at the margin**, i.e. interest saves taxes on the last dollar of income.

5.7 Which marginal tax rate should be used to compute after-tax cost of debt in case of a global company having operations in various countries?

The marginal tax rate for a global company will depend on where it decides to borrow money and not where it is incorporated. Generally, a global company would **borrow money where the tax rate is highest** (since that will yield the highest tax benefit from the interest deduction).

If, however, we can get information as to where the company borrows money in its annual report, we should use the marginal tax rate of the country in which the lending institution is located.

5.8 Can we consider accounts payables as debt?

You are correct, deferred tax liabilities are not considered just as accounts payables, etc. are not considered. And one reason we don't consider accounts payables as debt is because we can't explicitly see the interest rate associated with them (in the form of lost discounts). And it is messy to try and estimate an interest expense for them.

However, if you could explicitly see what is the interest cost of these liabilities, you can consider them as debt as well. These do get considered in our valuation in the form of lower operating income since their cost is included in the costs of doing business. For instance, let's assume that we are able to estimate an interest cost associated with accounts payable and treat that as debt. We would have to add back this expense and increase our operating income (the same way we don't reduce finance costs of debt when computing operating income). This increased operating income would result in a higher value of operating assets and then we would reduce the value of accounts payables come up with the value of equity.

Regarding your point of pension liabilities and contingent liabilities, you should definitely figure out the cost of those obligations (in the form of what would the company have to set aside to provide for those liabilities) and deduct from value of operating assets to come up with the value of equity.

6 DCF: Cost of capital

6.1 Why are market value weights used in computation of cost of capital?

The reason we use market values is because we take transaction driven view of discounted cash flow valuation, i.e. I would have to pay the market values in order to acquire this company today.

In essence, every discounted cash flow valuation is an acquisition valuation.

Some of the wrong reasons that you may come across for using market value weights are:

- **Markets are usually right:** If you assume this, and then go and value a company and come up with a different value for the company, the implicit assumption nullifies itself.
- **Market values are easy to obtain:** But so are book values. In fact, market value of debt may be harder to obtain than the book value of debt.
- **Book values are meaningless:** They may not reflect the true picture, but they are not meaningless.

6.2 Which value of debt should we deduct to come up with the value of equity - the market value of the debt or the book value of the debt?

Note that we use the market value of the debt to compute the debt ratio in the capital structure to compute the cost of capital. This is one part. The other part is, once you have discounted the cash flows to come up with the value of operating assets, which value of debt should we deduct to come up with the value of equity - the market value of the debt or the book value of the debt?

Now to answer that question, we need to distinguish between why we are doing the valuation - is it for liquidation purposes or for buying the company as a going concern.

- **Liquidation Value**

In this scenario, you should use the BOOK VALUE of debt to deduct to come up with the value of the equity. The reason is simple - When you liquidate the company, you are obligated to pay the "Book Value" of debt back to the lender and whatever remains would be distributed to the shareholders. The lender would not be satisfied with a lower payment, simply because you come up with a lower market value of debt.

- **Going Concern Value**

To understand why we should deduct the market value of debt from the value of operating assets to come up with the value of equity, it would help to understand why is it that we get a "lower/ higher" market value of debt compared to its book value. The reason for the divergence is the interest rate on THAT debt vs. the prevailing interest rates at which the company would be able to raise the debt if it were to raise it today.

If the interest rate on the debt you already have is HIGHER than the market interest rate, the market value of the debt will be higher than book value. This is the scenario where you say that the seller could argue that he is taking a haircut on his equity. And he should, simply because the buyer is going to be saddled with high interest-bearing debt when the market interest rates are low.

In real life, you wouldn't get into these problems as any rational seller would simply refinance (if he can) the debt obligations before getting into negotiations and the problem would be taken care of.

On the other hand, If the interest rate on the debt you already have is LOWER than the market interest rate, the market value of the debt will be lower than book value. This is the scenario where you say that the buyer could argue that he is paying a premium when he would be liable to pay back the book value of debt at maturity and not the market value. But think about it - the new buyer would also get to **enjoy the lower interest expenses for the life of the debt** than if he were to go and borrow from the market at present. As a seller of your business, that should be your argument since you were able to secure financing at a lower rate and you should be compensated for that.

In practical life, the negotiation would depend on who has the upper hand!

- **Special case of DISTRESS**

There is one special case, in which a company's debt is trading at much lower than the book value of the debt because of DISTRESS. In this scenario, if as a buyer, you (a healthy company) are acquiring control of this company, **you should use book value of debt since as soon as you acquire the company, its debt will start trading close to book value simply because now it has the backing of a healthy company and the default risk shrinks.** The debt was trading below book not because of the differential between interest rates, but because of the heightened default risk of the company.

6.3 Why use each year's cost of capital to discount cash flows?

The debate essentially is whether to discount the terminal value at the terminal value discount rate or to discount at each year's different discount rate separately. Here's my thoughts on the issue:

The cost of capital is an opportunity cost of investing in an asset that yields cash flows in the future. Inverting this problem and thinking in future value terms would help a great deal. Let's take a simple example with the following information – invest in a project which has a cash outflow of \$100 today and a one-time cash inflow of \$130 in year 5. Also suppose this asset is an extremely low risk asset which can be discounted at the risk-free rate. Our expectation of the risk-free rates over the next 5 years is as follows:

Year 1	8%
Year 2	7%
Year 3	6%
Year 4	5%
Year 5	4%

Now our opportunity cost is to invest the \$100 in risk free rates each year instead of investing in this project. In that case, we get a value of \$133.76 in year 5. Thus, comparing to the cash flow of \$130 of the project, we should not be investing in it.

However, if we use only the year 5 risk free rate to discount the cash flow of \$130 in year 5, we will end up overestimating the present value of that cash flow.

7 DCF: Cash flows

7.1 What are the different types of cash flows in valuation?

In valuation, we consider the following types of cash flows:

- **Free cash flow to firm (“FCFF”)**: These are cash flows which are available to all providers of capital to the firm after paying for all expenses (except interest) and investments. FCFF is computed as:

$$FCFF = EBIT * (1 - t) + Dep - Capex - \Delta WC$$

- **Free cash flow to equity (“FCFE”)**: These are cash flows that are potentially available to only equity holders of the firm and are computed as:

$$FCFE = NI + Dep - Capex - \Delta WC + Net\ debt\ issued - Preferred\ dividends$$

- **Dividends + Stock Buybacks**: These are actual cash flows that are distributed to the equity holders of the firm.

7.2 Why are tax savings from interest ignored when computing FCFF⁶?

The formula used for computation of FCFF ignores the tax savings from interest expenses. In effect, when computing FCFF, we pretend as if there is no debt in the firm and hence no interest expenses.

The reason for ignoring the tax savings from interest expenses is because these savings are considered in the **cost of capital** since we use the after-tax cost of debt when discounting cash flows. Considering these savings in the cash flows as well would amount to double counting.

7.3 Will increasing debt in the capital structure always a good thing since it increases free cash flow to equity?

It is true that as debt increases in a firm, the free cash flow to equity also increases. However, this may not necessarily increase value (i.e., it is not a free lunch), because of the following consequences:

- Increase in debt increases the beta and thereby the cost of equity. There will be an optimal debt ratio for a firm where the value is the highest. If debt is increased beyond this point, the increase in cash flows is more than offset by the increase in the cost of capital, thereby resulting in a lower value.
- Increase in debt beyond a certain point also affects the rating of the firm and consequently its cost of debt.

⁶ <https://www.capitalsurge.com/2018/01/15/why-are-tax-savings-from-interest-ignored-when-computing-free-cash-flow-to-firm/>

7.4 What are some of the adjustments that are needed to traditional accounting numbers when doing a valuation?

The following adjustments may be required to traditional accounting numbers:

- **Converting operating leases to debt:** Accounting classifies operating leases as expenses in the profit and loss statement. However, operating leases in reality are a form of debt since they fulfil the three criteria for classification as debt:
 - It's a contractual obligation;
 - It is tax deductible; and
 - Results in loss of control of the asset leased if commitments not honoured.

Following adjustments are needed in order to convert operating leases into debt:

Item	Adjustment
Liabilities	A debt liability equal to the Present Value of Operating Lease Commitments is created (discounted at pre-tax cost of debt)
Assets	A lease asset needs to be added to counter the liability of the same amount
After-tax Operating Income	$(\text{EBIT} + \text{Operating lease expense} - \text{Depreciation on leased asset}) * (1 - t)$
Note: First adjust then tax. This is because the total of imputed interest expense and depreciation on leased asset is set equal to the current year lease expense. i.e. Operating lease expense = Amortisation of lease expense + Imputed interest Thus, the above adjustment does not change the net income and hence does not alter the tax liability of the firm.	

- **Capitalising expenses** which are meant for creating benefits over more than 1 year (such as R&D, advertising, etc.) Following adjustments are required in order to capitalise expenses:

Item	Adjustment
Liabilities	The unamortized portion of R&D is added to the book value of equity
Assets	Create a research asset which increases the fixed assets by the amount of current year R&D expense

After-tax Operating Income	$\text{EBIT} \times (1 - t) + \text{Current year R\&D} - \text{Amortized portion}$ <i>Note that when R&D is declining, Current Year R&D expense < R&D Amortization and vice-versa</i>
Note: First tax then adjust. Since the entire current year R&D expense is deductible for tax purpose, we first deduct tax from pre-tax operating income and then adjust for R&D expenses.	

7.5 What type of expenses should be capitalised?

The key question whenever considering the capitalisation of expenses is to check whether that expense **creates benefits over multiple periods**. For customer acquisition, for instance, the question would be that once you have spent the money acquiring a new customer, does that customer stay with your business for more than 1 year. **If not, then these expenses need to be expensed.**

An interesting case study on this is Groupon, which used to add back customer acquisition costs to come up with an adjusted EBITDA no. to show to investors. The only problem was that there was no evidence of customers staying with Groupon once they stopped that expense.

Another thing to remember is that capitalising the expense **does not change cash flows or the discount rate**. So, a legitimate question is why should we capitalise the expenses in the first place and what impact does it really have on value? Capitalising changes the reinvestment rate which in turn changes the fundamental growth rate of earnings. And the question whether that adds to value or not would depend on how efficient that expense really is, whether your ROC from those expenses is more than the cost of capital.

7.6 Which method should be used to adjust for operating leases? Full adjustment or Imputed interest method?

Full adjustment where you compute the depreciation on the lease asset is better to compute operating income than the imputed interest method. This is because the implicit assumption in the imputed interest method is that the depreciation expense on the lease asset equates to the principal portion of the rental expense which is not interest. This is usually not the case and this would overstate the life of the lease asset. For example, when you use a pre-tax cost of debt of say 4%, the life of the asset is assumed to be 25 years (1/4%).

Example of Levi Strauss.

7.7 Should extension options on leases be considered as debt⁷?

Apart from capitalising minimum lease commitments, the new lease accounting standard also capitalises extension options. The question then becomes whether it is appropriate to consider

⁷ See email exchange dated 5-Mar-21.

these extension options as debt? On the one hand, these are not contractual commitments as they are only options and therefore, should not be considered as debt. On the other hand, if we think the company is going to continue to act as a going concern, the probability of exercising these options is likely extremely high.

The right answer is that extension options SHOULD NOT BE CONSIDERED AS DEBT. Here is what Prof. Damodaran has to say about the issue:

“The reason we include leases in debt is not because we just expect them to continue but because they are contractual commitments. Once you incorporate that lease debt into your debt ratio, and keep that debt ratio fixed, you are already assuming that lease debt will not just be renewed but grow as your operations grow.”

This is a rule we follow with all debt. Thus, with a growing manufacturing that we expect will be borrowing money in the future to build more factories, we still stick with the existing debt, when computing the cost of capital. The reason we do so is simple. Once debt is built into the debt ratio and used to compute a cost of capital, and we use that cost of capital to discount future cash flows, we are implicitly. Growing the dollar debt at the same rate as the firm is growing. In effect, we are already building in the expected increase in debt, through the use of a debt ratio in the cost of capital.

7.8 How should we deal with variable lease costs?

Variable lease costs are lease costs that are contingent on an event happening. For example, many retailers have two components to their lease costs:

- A minimum amount that they have to pay fixed component; and
- A variable component which is based

7.9 What are the types of taxes used in valuation?

There are two types of taxes rates which are relevant for valuation:

- **Marginal tax rate:** The tax rate on the last dollar of income;
- **Effective tax rate:** The average tax that is paid by the company on its taxable income (i.e. Pre-tax income).

Usually, Marginal tax rate is > Effective tax rate. This is generally because of any of the following:

- Foreign income which might be taxed at a lower rate;
- Special incentives in the tax law; and
- Tax deferral strategies.

7.10 What tax rate should be used in computing after-tax operating income?

We should start the effective tax rate and gradually move towards the marginal tax rate in later years. The reason for this is that if we use the effective tax rate forever, we would implicitly be assuming that the firm can defer taxes forever, which is not a prudent assumption.

7.11 What is the best way of dealing with net operating losses?

Some analysts value the company as if there are no losses carried forward. They then separately value the net operating losses simply by multiplying the NOL with the marginal tax rate and adding it to the value obtained earlier. However, this approach has two drawbacks:

- It does not consider the time value of money since NOLs can be offset only against operating income and operating in a particular year may not be sufficient to absorb the entire carried forward NOL; and
- Secondly, in case of a money losing company, it is not even certain whether the firm will even get the benefit of the NOL since they can be forward only for a certain number of years before they lapse.

The best way then, would be not to keep operating losses separate but bring them into the valuation. Keep track of the NOLs that the firm is carrying forward and project income based on them. For instance, give the benefit of the operating losses carried forward in years when the income is positive.

7.12 What is the linkage between capital expenditures and growth?

Net capex is generally a function of how fast a firm is growing or expecting to grow. Therefore, assumptions about net capex can never be made independently of assumptions about growth in the future.

7.13 Should acquisitions through the issuing shares of the acquiring company be considered as Capex?

Acquisitions, whether through cash or through issue of shares should be considered as Capex. Acquiring through the issue of shares essentially skips one step – instead of raise money through shares and spend that money for the acquisition, you have simply given the shares away.

7.14 What is the definition of working capital that is useful for valuation purposes?

In accounting terms, working capital is measured as,

$$\text{Working capital} = \text{Current assets} - \text{Current liabilities}$$

where, current assets include cash and current liabilities include short term debt and current portion of long-term debt.

However, **for valuation purposes**, we use the following definition:

$$\text{Working capital} = \text{Non cash current assets} - \text{Non debt current liabilities}$$

7.15 Why is cash and short-term debt excluded from computation of working capital for valuation purposes?

Cash: The reason we exclude cash is because it is generally not a wasting asset. Inventories, debtors, etc. do not earn any return and that is why they are included in working capital. Cash earns a money market rate of return (which is a fair return for a guaranteed cash flow) and hence is not included. If, however, cash does not earn anything, then it should be included in working capital.

Short-term debt: The reason for not including short term debt is that it is included in debt.

7.16 Why have companies started returning capital in the form of buy-backs than through dividends?

Over the last few decades, especially since the 1990s, companies have increasingly returned capital more in the form of buy-backs than in the form of dividends. One very plausible, but flawed explanation that is often given is the differential tax treatment of buy-backs and dividends – buy-backs being taxed at a lower rate than dividends which are taxed as ordinary income. If this explanation was correct, then companies would have bought back stock more in the 1960s as the differential in tax rate was much higher in the 1960s than it has been beginning the 1990s. Thus, the correct explanation has to do more with a change in the 1990s.

- **Employee stock options:** 1990s was a year when companies started compensating employees a lot more through employee stock options than through cash. Thus, employees' are worth more, higher the share price of the company. Dividends have the effect of lowering the price per share of the company once they are paid. Buy-backs, on the other hand, have the opposite effect, i.e., they increase the price of the security. Thus, as managements have started getting compensated more through stock options, they have increasingly resorted to buy-bacs as opposed to dividends purely out of self interest than anything to do with rationality.
- **Hooking up:** Another possible explanation is that dividends are seen as sticky. Once a company starts paying a certain dividend, it feels obligated to continue that dividend policy even though it may not make sense for the company. For instance, data from the US suggests that even in recession years, more companies increase dividends than those who decrease them. Also, dividends tend to lag earnings, i.e., companies generally tend to wait for growth over a number of years to pass on increases in dividends to shareholders.

Buy-backs, on the other hand, are considered irregular and hence, do not obligate a company to return capital through a regular buy-back program. They provide a lot more flexibility to managements.

In a nutshell, dividends are like GETTING MARRIED whereas, Buy-backs are like HOOKING UP!

8 DCF: Growth

8.1 When to use $(1 + g)$ and when not to?

If you are given next year's number (earnings, return on capital, margin etc.), you don't need a $(1+g)$ in the numerator. If you are given the numbers for the most recent year, you do need a $(1+g)$ in the numerator. Thus, if the question mentions that the expected operating income or margin or return on capital next year is X, you do not need to put the $(1+g)$ in the model.

If nothing is mentioned, Prof. Damodaran usually assumes they are next year numbers. However, you are free to assume differently. On the exam, it would be mentioned clearly.

8.2 Most people consider that growth leads to higher value for a business. Is this true?

The general misconception is that higher growth leads to higher value. However, this is not true. Growth is a double-edged sword, i.e., it has a good side as well as a bad side:

The good side	The bad side
Pushes up revenues and operating income	Need to set aside money to create growth – reinvestments which lower cash flows
The net effect of growth is whether the good outweighs the bad!	

8.3 What are the ways in estimating growth rates for a firm?

There are three ways of estimating future growth:

Look at the PAST	Look at OTHERS	Look at FUNDAMENTALS
The historical growth in EPS is a good starting point.	Analysts estimate growth in EPS for many firms. It is useful to know what their estimates are.	Ultimately, all growth in earnings can be traced to two fundamentals: • How much the firm is investing in new projects; and • What returns these projects are making for the firm

Growth rates are one of the most important numbers in a valuation and hence, **should not be outsourced to others.**

8.4 What are some of the problems with historical growth rates?

Historical growth rates suffer from the following problems:

- Can be estimated in several different ways:
 - Arithmetic vs. Geometric averages.
 - Simple vs. Regression models.
- Historical growth rates can be sensitive to:

- The period used in the estimation (starting and ending points).
- The metric that the growth is estimated in – revenues vs. operating income vs. EPS.
- You must wrestle with the following:
 - How to deal with negative earnings.
 - Effects of scaling up.

8.5 What are some of the problems with analyst growth rates?

- Analysts estimate **growth in EPS**. However, for the purposes of a DCF valuation, it is best to have a growth in revenues or operating income.
- Analyst growth rates are generally **highly correlated with historical growth rates** – which may or may not be a good estimate of future growth.

8.6 What are some of the biases that analysts may suffer from when predicting growth?

Following is a list of biases that analysts may suffer from when covering a company:

- **Tunnel Vision:** Becoming too focused on the sector and valuations within the sector that they lose sight of the bigger picture.
- **Lemmingitis:** Analysts may experience strong urge to change recommendations and revise earnings estimates when other analysts do the same.
- **Stockholm Syndrome:** Analysts who start identifying with the managers of firms they are supposed to follow.
- **Factophobia:** Tendency to base recommendations on a “story” coupled with a refusal to face facts.
- **Dr. Jeckyll/ Mr. Hyde:** Analysts who think their primary job is to bring investment banking business to the firm.

8.7 What are some of the propositions we can make about analyst growth rates?

- There is far less private information and far more public information in most analyst forecasts than is generally claimed.
- The biggest source of private information for analysts remains the company itself which might explain:
 - Why there are more buy recommendations than sell recommendations (information bias and need to preserve sources);
 - Why there is such a high correlation across analyst forecasts and revisions; and
 - Why All-America analysts become better forecasters than other analysts after they are chosen to be part of the team.
- There is value in knowing what analysts are forecasting as earnings growth for a firm. There is danger, however, when they agree too much (lemmingitis) and when they agree to little (in which case the information that they have is so noisy as to be useless).

8.8 What is the fundamental growth equation?

Growth in earnings can be computed as follows:

Expected growth = Growth from new investments + Efficiency growth

*Growth from new investments = Reinvestment Rate * RoC_{t+1}*

$$\text{Efficiency growth} = \frac{RoC_{t+1} - RoC_t}{RoC_t}$$

8.9 How is growth from new investments derived?

As mentioned above, there are two components to growth from new investments:

- Reinvestment rate; and
- Return on capital.

The derivation of this formula is explained with the help of the following example:

Investment in Existing Projects (\$1000) X Current RoI (12%)				Current Earnings (\$120)
Investment in Existing Projects (\$1000) X Next Period's RoI (12%)	+	Investment in New Projects (\$100) X RoI on New Projects (12%)	=	Next Period's Earnings (\$132)
Investment in Existing Projects (\$1000) X Δ RoI current to next period (0%)	+	Investment in New Projects (\$100) X RoI on New Projects (12%)	=	Change in Earnings (\$12)

Special case: RoI on existing projects remains unchanged and is equal to RoI on new projects

Investment in New Projects	X	Return on Investment	=	Change in Earnings
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$\frac{(\$100)}{\text{Current Earnings } (\$120)}$		(12%)		$\frac{(\$12)}{\text{Current Earnings } (\$120)}$
Reinvestment Rate (83.33%)	X	Return on Investment (12%)	=	Growth rate in Earnings (10%)

General case: RoI can change from period to period		
$\frac{\text{Investment in [Existing Projects* (Change in RoI) + New Projects*RoI]}}{\text{Investment in Existing Projects*Current RoI}}$	=	$\frac{\text{Change in Earnings}}{\text{Current Earnings}}$
$\frac{\$1000*(13\% - 12\%) + \$100*13\%}{\$1000*12\%} = \frac{\$23}{\$120}$	=	19.17%

8.10 Which measures of reinvestment and return should be used when trying to compute the growth rate?

Earning measure	Reinvestment measure	Return measure
Earnings per share	$\begin{aligned} &\text{Retention ratio}^8 \\ &= \\ &\% \text{ of net income retained by the company} \\ &= \\ &1 - \text{Payout Ratio} \end{aligned}$	$\begin{aligned} &\text{Return on Equity} \\ &= \\ &\frac{\text{Net Income}}{\text{Book Value of Equity}} \end{aligned}$
Net income from non-cash assets	$\begin{aligned} &\text{Equity reinvestment rate} \\ &= \\ &\frac{(\text{Net Capex} + \text{Change in non-cash WC} - \text{Change in Debt})}{\text{Net income from non-cash assets}} \end{aligned}$	$\begin{aligned} &\text{Non-cash RoE} \\ &= \\ &\frac{\text{Net income from non-cash assets}}{(\text{Book Value of Equity} - \text{Cash})} \end{aligned}$

⁸ This assumes that the all earnings retained are reinvested into the business –this may not be true. The proposition is that the expected growth rate in earnings cannot exceed the RoE in the long term.

	Net Income	
	Reinvestment rate	Return on capital or ROIC
	=	=
Operating income	(Net Capex + Change in Non-cash WC)	After-tax operating income
	÷	÷
	EBIT*(1 – t)	(Book value of equity + Book value of debt – Cash)

8.11 What is efficiency growth?

Earnings from existing projects can grow **only if return on investment increases on existing projects**. When RoC is changing, there will be a second component to growth, positive if the RoC is increasing and negative if the RoC is decreasing.

If RoC_t is the return on capital in period 't' and RoC_{t+1} is the return on capital in period 't+1', the expected growth rate in operating income will be:

$$\text{Expected growth} = \text{Reinvestment} * RoC + \frac{RoC_{t+1} - RoC_t}{RoC_t}$$

The second half of the equation (after the '+' sign) is the growth from efficiency. If the change is over multiple periods, the second component should be spread out over each period.

More scope in companies which have very low returns on capital. **However, this is a finite way to grow because after a while you can only squeeze efficiency so much.**

Proposition

In the long term, you cannot grow if you do not reinvest (i.e., grow your asset base)

8.12 Can stable growth be negative?

There is no reason why stable growth cannot be negative since the terminal value can still be estimated. Intuitively, a negative stable growth rate implies that the firm is partially liquidating itself (making itself smaller) each year until it just about disappears.

In fact, this **may be the right choice** to make when valuing firms in industries that are being phased out because of:

- Technological advances (manufacturers of land line phones with the advent of cell phones);
- Where a critical customer is scaling back purchases for the long term (defence contractors after the end of the cold war.)

8.13 What are some of the key assumptions an analyst needs to make on stable growth?

There are three critical assumptions that one needs to make about stable growth in any discounted cash flow valuation:

- When will the firm become a stable growth firm, if it is not already one?
- What characteristics the firm will have in stable growth – assumptions need to be made about return on incremental investments, cost of equity/ capital?
- How will the firm make a transition from high growth to stable growth?

8.14 Length of the high growth period

8.15 Session 9 Question 6 and 7

"I have been noticing the debate about the terminal value calculation in the session 9, questions 6 & 7 with interest. I could have intervened earlier but I wanted to see how it would play out. Let me lay out what Mark noted in trying to solve the problem (in red below)

Questions 6 and 7 start with operating income growth of 7% a year for 3 years, and then explore terminal values based on growth rates thereafter of 3%, 0%, and -5%.

The model answers to both questions start with this calculation:

Expected after-tax operating income in year 4 = $40 (1.07)^3(1.03) = \$50.96$

This bakes into they year 4 after-tax operating income a growth of 3% in year 4. This makes sense for the 3% growth scenario, but it doesn't make sense for the 0% and -5% scenarios, and yet this \$50.96 year 4 number is used in the model answer for all of the scenarios.

Question 6 is interesting in that I expect it to show there is no change in value when the growth rate changes given a return on capital equal to the cost of capital -- but I don't get that result. Changing the year 4 growth to match the perpetual growth, I arrive at different values, thus:

$$TV = 40 * 1.07^3 * 1.03 * (1 - 0.03/0.1) / (0.1 - 0.03) = 35.33 / 0.07 = 504.7$$

$$TV = 40 * 1.07^3 * (1 - 0) / 0.1 = 490$$

$$TV = 40 * 1.07^3 * 0.95 * (1 - (-0.05/0.1)) / (0.1 + 0.05) = 46.552 * 1.5 / 0.15 = 465.5$$

The difference in each value is only the difference in that one year of growth.

$$490 * 1.03 = 504.7$$

$$490 * 0.95 = 465.5$$

Not sure how best to untangle this.

As he notes, the logic that growth should not matter when ROIC = Cost of capital is strong, but the math seems to suggest that it matters, even when ROIC = Cost of capital. The reason actually lies in how we (or at least I) have been computing growth rates so far in this class.

Each year, we assumed that growth rate in year t = Reinvestment Rate in year t * ROIC in year t . However, there is a timing problem here, since your reinvestment in year t actually happens at the end of year t , making it impossible to deliver growth in that year. Here is what it should actually be:

Growth rate in year t = Reinvestment rate in year $t-1$ * ROIC in year t

Consequently, you will get the growth of 3% in your terminal year in all your stable growth rate scenarios, since the reinvestment for it was made in year before. Thus, your cash flow in year 4 will always be $=40 * 1.07^3 * 1.03$ (which as Sam noted should be 50.47, not 50.96, which was a sloppy error on my part), no matter what your stable growth rate is, thus making your terminal value invariant.

You may wonder why I have been sloppy and not tied the growth rate to the previous year's reinvestment rate. It is a pain in the neck to do and it can cause considerable heart ache in complex companies and the effect on valuation is generally trivial. So, if you want to be precise from this day on, model your reinvestment and growth assumptions with a lag. I'll continue to be lazy. And for all of you who wrestled with this question and feel frustrated that I did not jump in with the answer earlier, my defense is that this is the way you master the craft!"

On a different note, I hope that some of you had a chance to read Shashank's retransmission of a note that I sent the other class about the lag between reinvestment and growth. To make our lives simple, we assume that reinvestment and growth happen simultaneously. Thus, if you reinvest 60% of your earnings and have a return on capital of 20%, we assume that the growth rate will be 12% that year. In reality, since reinvestment occurs at the end of each year, the growth rate will lag the reinvestment by a year. Consequently, in the year just before your terminal year, your reinvestment rate should drop to your stable period levels. To the question of why we don't do this right, the answer is that the effect on valuation, most times, is small enough to gloss over.

8.16 How to easily compute the mid-year convention?

One of the simplest ways of computing the mid-year convention is to take whatever value you get and simply move it forward by half a year at the cost of capital to get the mid-year convention without any of the pain. However, it is better to remember that the mid-year convention adds little to the analysis and thus, spending as little time on it is preferable.

8.17 Reinvestment = Invested capital

Balance Sheet: Year 0			
Assets	\$	Liabilities	\$
Cash	100	Payables	40
Receivables and Inventory	70	Debt	100
PP&E	100	Equity	130
Total	270	Total	270

Balance Sheet: Year 1			
Assets	\$	Liabilities	\$
Cash	110	Payables	50
Receivables and Inventory	90	Debt	110
PP&E	110	Equity	150
Total	310	Total	310

9 DCF: Terminal Value

9.1 Why is the growth in perpetuity restricted to the risk-free rate of the economy?

The argument for restricting the long-term growth of a company to the risk-free rate of the economy is rooted in economic fundamentals. The risk-free rate is composed of expected inflation and the expected real interest rate. i.e.,

$$R_f = \text{Expected Inflation} + \text{Expected Real Interest Rate}$$

where, the real interest rate is what the borrowers agree to return to lenders in real goods/ services. If this does not hold true in any reasonable period of time, lenders will not be incentivised to lend.

Similarly, the nominal GDP growth can be written as:

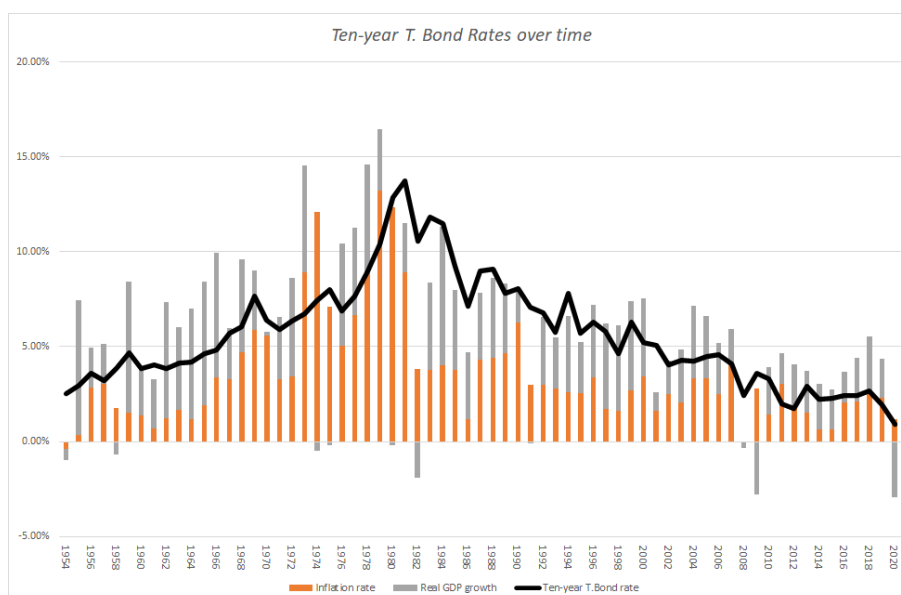
$$\text{Nominal GDP growth} = \text{Expected Inflation} + \text{Expected Real Growth}$$

where, the real growth rate in the economy measures the expected growth in the production of goods and services.

However, the argument for risk-free rate being equal to the nominal GDP growth has two legs to it:

- In the long term, the real growth rate cannot be lower than the real interest rate since the growth in goods/ services must be enough to cover the promised rate.
- In the long term, the real growth rate can be higher than the real interest rate, to compensate risk taking. However, as economies mature, the difference should get smaller and **since there will be growth companies in the economy, it is prudent to assume that the extra growth comes from these companies.**

Although we can never purely test whether this hypothesis is correct since, expected inflation and expected growth represent expectations and can never be actually observed. However, we can make some reasonable conclusions by observing whether this relationship has held up to historically.



An alternate explanation:

In my view, there is an alternate explanation as to why this relation should hold in the long term. Since we are assuming the risk-free rate as the long-term government bond rate (that does not have any default risk in it), we are essentially dealing with two asset classes:

- Debt
- Equity

Now suppose that long term expectations of growth in earnings of the economy is much higher than the long-term government rate. In this scenario, for some time, equity shareholders will end up borrowing money and investing to earn much higher returns - Looking to this, people investing in bonds, not being satisfied with their returns, will be attracted to investing in equities instead of bonds. This would have the impact of lowering bond prices and therefore, increasing prospective bond returns to a level which adequately compensates the bondholders for growth in the real economy.

Of course, you are unlikely to reach a stage of pure equilibrium at any given point in time but, this should be true on average over the long term.

10 DCF: Loose ends in valuation

10.1 When should assets be added separately to the valuation you have come up with?

Usually, assets on the balance sheet of the company should be added separately only if they are not essential to the operations of the company. For instance, below are some assets and how they should be dealt with in a valuation:

- **Real estate:** The value of real estate **should not be included** if it is being used in the operations of the company.
- **Vacant land:** If it is not essential to the operations of the company then you **should add** the value of the vacant land to the value of the company.
- **Patents:** Your revenues will already be reflecting the benefits of the patent and hence is **already included** in the value.
- **Brand name:** The value of brand names should be included in the revenues and margins of the company. Accordingly, **should not be added** separately.
- **Goodwill:** It is a plug variable and not even an asset; and

10.2 How should we incorporate the value of cash in a valuation?

The simplest way to deal with cash is to **keep it out of the valuation** and bring it into the picture once the operating assets have been valued. Accordingly, the cash flows should be:

- Before interest income from cash and marketable securities; and
- Discount rate should not be contaminated by inclusion of cash.

This approach will allow you to value the operating assets of the firm. Once the operating assets have been valued, we simply need to add the book value of cash and marketable securities.

One of the main reasons for using this approach is this – If we include the interest income from cash in the cash flows, we would also need to adjust the discount rate for the presence of cash, i.e. Beta should be weighed down by the cash holdings. Unless cash remains a fixed percentage of value over time, these valuations will tend to break down.

10.3 When should cash be discounted or valued at a premium?

Sometimes, it may make sense to discount the cash or value it at a premium. However, contrary to popular belief⁹, this has nothing to do with the presence of cash but with the fear of what managers will do with that cash. The scenarios in which cash should be valued at other than its book value are:

⁹ Some argue that cash should be discounted simply because it earns a lower rate of return. This argument is flawed because it violates one of the first principles of valuation and corporate finance – the discount rate should reflect the riskiness of the cash flows. In the case of cash, the cash flows generated by cash are risk free and hence, as long as cash is earning a risk-free rate of return, it should not be discounted at the normal cost of capital for the company's operating cash flows.

Discount Cash if	Firm's managers have a history of taking projects which earn less than their cost of capital. This is basically a stupidity discount being applied to the cash of the firm. <i>It is not the cash that worries investors but what the managers will do with that cash if they don't give it back to shareholders.</i>
Value Cash at a premium if	• Capital markets have a tendency to shut down; and • The firm has a history of taking projects which earn a return higher than its cost of capital. <i>When capital markets become inaccessible, cash becomes a strategic weapon.</i>

10.4 Should there be a discount for complexity, i.e., should two companies with exactly the same cash flows be valued differently because one is more complex than the other?

The traditional view has been to allow for no complexity discount because of diversification – meaning that as long as an investor is well diversified, the good and bad news will cut in both directions and thus, will be neutralised.

However, the flaw in the above logic is that what you don't know is what managers of these companies have CHOSEN NOT TO TELL you. And generally, managers withhold bad news and not good news. So, complexity is likely to cut in the negative direction only.

Therefore, complex companies should trade at a discount to simple companies EVEN IF EVERY NO. LOOKS THE SAME.

Acquisition Valuations

11 Acquisition valuation

11.1 Who wins in an acquisition?

Studies have shown that acquisitions are great for the target companies but not always for the acquiring company stockholders.

Proposition

You should try to be the target than the acquirer in an acquisition because the target appears to consistently win (apart from the bankers to the deal of course!)

11.2 The long-term follow-up of acquisitions

Managers often argue that the market is unable to see the long-term benefits of mergers that they can see at the time of the deal. If they are right, mergers should create long term benefits to acquiring firms.

However, the evidence¹⁰ does not support this hypothesis. In fact, many acquisitions are reversed within a fairly short period of time.

The underlying theme in acquisitions is that growing through acquisitions seems to be a loser's game.

Firms that grow through acquisitions have generally had far more trouble creating value than firms that grow through internal investments. In general, acquiring firms tend to:

- Pay too much for target firms.
- Overestimate the value of “synergy” and “control”.
- Have a difficult time delivering the promised results.

Worse still, there seems to be very little learning built into the process. The same mistakes are made over and over again, often by the same firms with the same advisors.

Conclusion

There is structurally wrong with the process for acquisitions which is feeding into the mistakes.

11.3 What are the seven sins of acquisitions

1. Risk transference: Attributing acquiring company risk characteristics to the target firm.

¹⁰ Studies by McKinsey and Co., KPMG

2. Debt subsidies: Subsidizing target firm shareholders for strengths of the acquiring firm.
3. Auto-pilot control: The “20% control premium” and other myths.
4. Elusive synergy: Misidentifying and mis-valuing synergy.
5. It’s all relative: Transaction multiples, exit multiples, etc.
6. Verdict first, trial afterwards: Price first, valuation to follow.
7. It’s not my fault syndrome: Holding no one accountable for delivering results of an acquisition.

We'll test each of these with the help of an example:

Particulars	\$ millions
Revenues	100
- Operating expenses	80
Operating income	20
Taxes	8
After tax operating income	12
Assume that this firm has no debt outstanding and will generate this operating income forever (with no growth) and that the cost of equity is for this firm is 20%	
Value of the company ($12 / 0.20$)	60

11.4 Test 1: Risk transference

Assume that as an acquiring firm, you are in a much safer business and have a cost of equity of 10%. What is the value of the target firm to you? In other words, would you use a cost of capital of 10% or 20% when valuing this company as a target of an acquisition?

Lesson 1: Don't transfer your risk characteristics to the target firm

- The cost of equity used for an investment should reflect the risk of the investment and not the risk characteristics of the investor who raised the funds.
- Risky businesses cannot become safe just because the buyer of these businesses is in a safe business.

e.g. AT&T in late 1990s

11.5 Test 2: Cheap debt

Assume as an acquirer that you have access to cheap debt (at 4%) and that you plan to fund half the acquisition with debt. How much would you be willing to pay for the target firm. In other

words, when you compute your cost of capital, would you use the cost of debt at which you can raise funds or the cost of debt which applies to this particular investment?

Lesson 2: Render unto the target firm that which is the target firm's but not a penny more

- As an acquiring firm, it is entirely possible that you can borrow much more than the target firm can on its own and at a much lower rate. If you build these characteristics into the valuation of the target firm, you are essentially transferring wealth from your firm's stockholders to the target firm's stockholders.
- When valuing a target firm, use a cost of capital that reflects the debt capacity and cost of debt that would apply to that firm.

e.g. GE

11.6 Test 3: Control premiums

- Assume that you are now told that it is conventional to pay a 20% premium for control in acquisitions (backed up by Mergerstat). How much would you be willing to pay for the target firm?
- Would your answer change if I told you that you can run the target firm better and that if you do, you will be able to generate a 30% pre-tax operating margin (rather than the 20% margin that is currently being earned).
- What if the target firm were perfectly run?

Lesson 3: Beware of rules of thumb

- Valuation is cluttered with rules of thumb. It is common practice to add arbitrary premiums for brand name, quality of management, control, etc.
- If you have done your valuation right, those premiums should already be incorporated in your estimated value. Paying a premium will lead to double counting.

11.7 Test 4: Synergy

A few questions to ponder:

- Assume that you are told that the combined firm will be less risky than the two individual firms and that it should have a lower cost of capital (and a higher value). Is this likely?
- Assume now that you are told that there are potential growth and cost savings synergies in the acquisition. Would that increase the value of the target firm?
- Should you pay this as a premium?

11.7.1 What is synergy

Synergy reflects the value of things that you can do as a combined company **THAT YOU COULD NOT HAVE DONE** as standalone company. Synergy requires two companies.

11.7.2 Combining two companies makes them safer but does not necessarily add value to the company. How do you reconcile this?

This is because when we value individual companies, we consider risk that we cannot diversify away. Accordingly, the risk which gets lowered by combining the companies is diversifiable risk which does not enter the valuation in the first place.

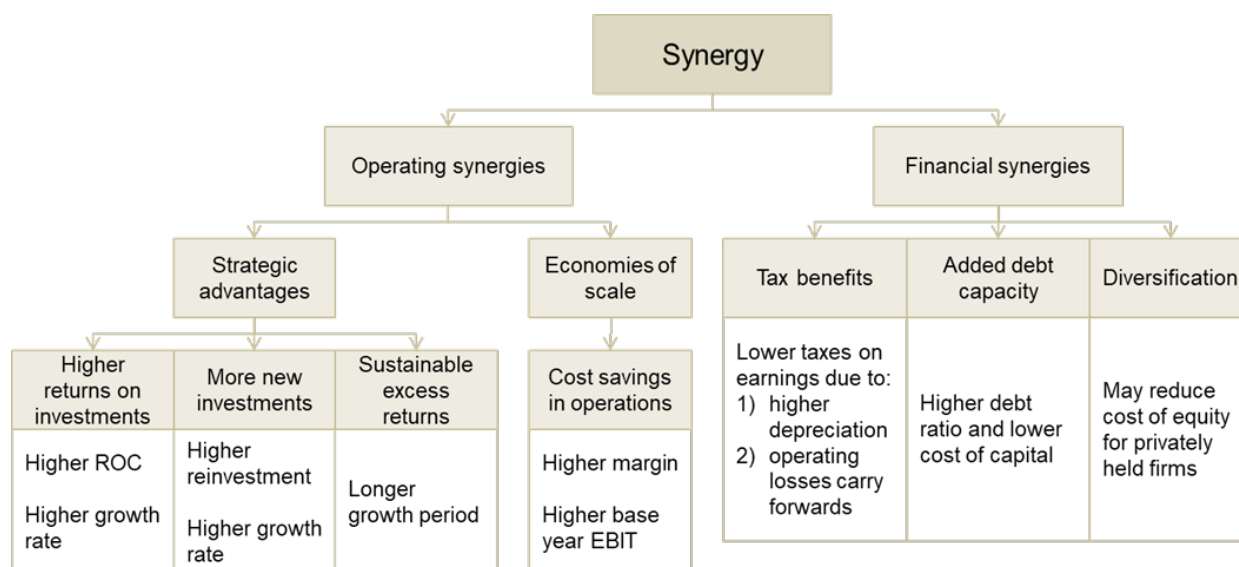
The intuition is this:

If a steel company buys a chemical company and tells its stockholders that it has created value because it is now safer, your response as a stockholder should be that I could have bought the chemical company myself in my portfolio and I would not have paid a 30% premium to do it!

The exception to this rule is when two private companies combine with each other to create a safer company because we look at total risk in such cases.

11.7.3 The value of synergy

There are different types of synergies that can create value in a merger/ acquisition. Some of these have been depicted below:



11.7.4 Valuing synergy

The value of synergy can be estimated in three steps:

Step 1	Step 2	Step 3
--------	--------	--------

Value the firms involved in the merger independently	Value the combined firm with no synergies	Effects of synergy are built into expected growth and cash flows and the combined firm is re-valued with synergy ¹¹
Use cost of capital of the specific firm	Simply adding the values obtained from Step 1	Value of synergy = Value of combined firm with synergy (–) Value of combined firm w/o synergy

11.8 Test 5: Comparables and exit multiples

Now assume that you are told that an analysis of other acquisitions reveals that acquirers have been willing to pay 5 times EBIT. Given that your target firm has an EBIT of \$20 million, would you be willing to pay \$100 million for the acquisition?

What if I estimate the terminal value using an exit multiple of 5 times EBIT?

As an additional input, your investment banker tells you that the acquisition is accretive¹²

11.8.1 Biased samples lead to poor results

- Biased samples yield biased results. Basing what you pay on what other acquirers have paid is a recipe for disaster. After all, we know that acquirers, on average, pay too much for acquisitions. By matching their prices, we risk replicating their mistakes;
- Even when we use the pricing metrics of other firms in the sector, we may be basing the prices we pay on firms that are not truly comparable; and
- When we use exit multiples, we are assuming that what the market is paying for comparable companies today is what it will continue to pay in the future.

Lesson 5: Don't be a lemming

- All too often, acquisitions are justified by using one of the following two arguments:
 - Everyone else in your sector is doing acquisitions. You have to do the same to survive;
 - The value of a target firm is based upon what others have paid on acquisitions, which may be much higher than what your estimate of value for the firm is.
- With the right set of comparable firms, you can justify almost any price

¹¹ If you must wait for some time (say 3 years) for synergies to take effect then, you should discount the value of synergy for 3 years at the **combined company's cost of capital**.

¹² Your PE ratio is 20 whereas the PE ratio of the target is only 10, therefore, you will get a jump in earnings per share after the acquisition.

- EPS accretion is a meaningless measure. After all, buying a company with a PE lower than yours will mathematically lead to EPS accretion no matter what the value of the company.

11.8.2 Test 6: The CEO really wants to do this... or there are competitive pressures...

Now assume that you know that the CEO of the acquiring firm really, really wants to do this acquisition and that the investment bankers on both sides have produced fairness opinions that indicate the firm is worth \$100 million. Would you be willing to go along with this?

Now assume that you are told that your competitors are all doing acquisitions and that if you don't do them, you will be at a disadvantage. Would you be willing to go along?

Lesson 6: Don't let egos or investment bankers get the better of common sense

- If you define your objective in a bidding war as winning the auction at any cost, you will win. But beware the winner's curse.
- The premiums paid on acquisitions often have nothing to do with synergy, control or strategic considerations (though they may be provided as the reasons). They may just reflect the egos of the CEOs of the acquiring firms. There is evidence that "overconfident" CEOs are more likely to make acquisitions and that they leave a trail across the firms that they run.
- Pre-emptive or defensive acquisitions, where you over pay, either because everyone else is overpaying or because you are afraid that you will be left behind if you don't acquire are dangerous. If the only way you can stay competitive in a business is by making bad investments, it may be best to think about getting out of the business.

11.9 Test 7: Is it hopeless

The odds seem to be clearly weighted against success in acquisitions. However, if you were to create a strategy to grow, based upon acquisitions, which of the following offers you best chance of success:

This	Or this	What is better (template)
Sole bidder	Bidding war	Sole bidder
Public target	Private target	Private targets
Pay with cash	Pay with stock	Tricky ¹³ Cash for large deals Stock for small deals (if stock is overvalued)

¹³ If it is a small deal, you may be better off paying with stock. For a large deal, it is mostly better to pay with cash.

Small target	Large target	Small target
Cost synergies	Growth synergies	Cost synergies

- Studies that have focused on synergies have concluded that you are far more likely to deliver cost synergies than growth synergies.
- Synergies that are concrete and planned for at the time of the merger are more likely to be delivered than fuzzy synergies.
- Synergy is much more likely to show up when someone is held accountable for delivering synergy.
- You are more likely to get a share of the synergy gains in an acquisition when you are a single bidder than if you are one of multiple bidders.

Lesson 7: For acquisitions to create value, you have to stay disciplined

- **If you have a successful acquisition strategy, stay focused on that strategy. Don't let size or hubris drive you to "expand" the strategy.**
- **Realistic plans for delivering synergy and control have to be put in place before the merger is completed. By realistic, we have to mean that the magnitude of the benefits have to be reachable and not pipe dreams and that the time frame should reflect the reality that it takes a while for two organizations to work as one.**
- **The best thing to do in a bidding war is to drop out.**
- **Someone (preferably the person pushing hardest for the merger) should be held to account for delivering the benefits.**
- The compensation for investment bankers and others involved in the deal should be tied to how well the deal works rather than for getting the deal done.

Relative Valuation

12 Relative Valuation (Introduction)

12.1 Information requirements

An analyst says that he never does a DCF valuation because it requires too many assumptions. He argues that it is far simpler to use a multiple obtained by looking at other firms in the sector, to estimate value. Is he right?

No. Even when you are using pricing, you are implicitly making assumptions about the same variables as in a discounted cash flow valuation. The only difference is that in a DCF, those assumptions are explicit.

12.2 Distributional assumptions

If you estimate the PE ratio for all companies and graph out the frequency distribution, can the distribution be normal?

No, because a normal distribution ranges from negative infinity to positive infinity. PE ratios cannot be negative. Negative PE ratios are not meaningful.

12.3 Controlling variables

You are trying to decide whether a software company is fairly priced, based upon its PE ratio. The company trades at a PE ratio of 12 and the average for the software sector is 20. Based on this comparison, what would you conclude:

- The stock is cheap;
- The stock is expensive; or
- The stock is fairly priced

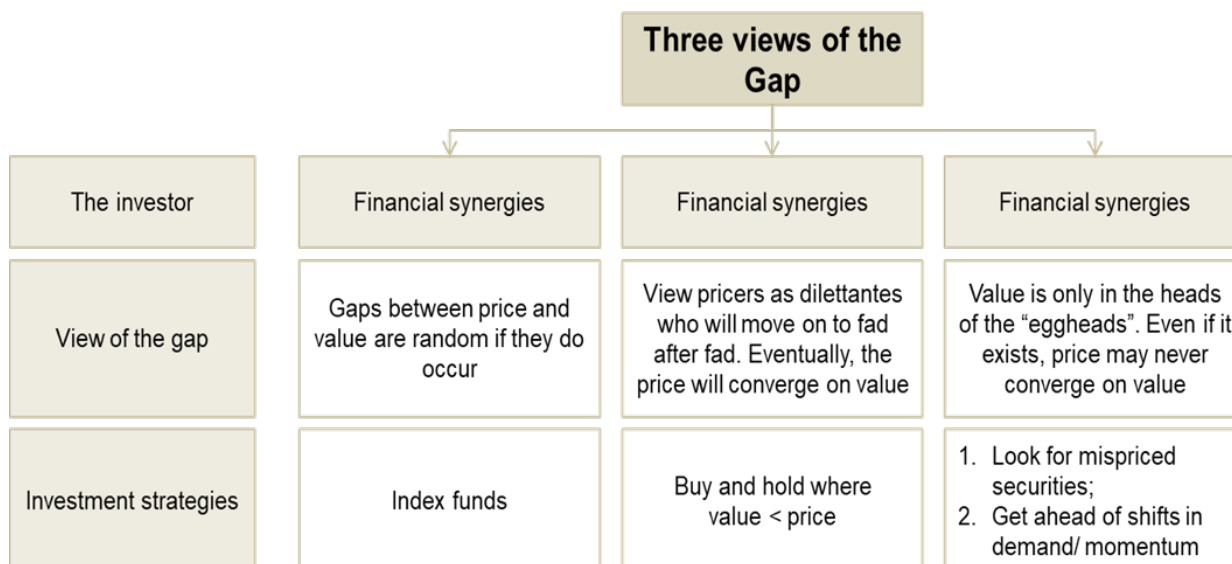
Could be anything, because of differences in growth, risk and cash flows relative to the sector. 95% of the stocks that look cheap deserve to be cheap.

The essence of relative valuation is to find the 5% that are actually cheap.

12.4 Value vs. Price

Determinants of Value	Determinants of Price
1. Cash flows from existing assets; 2. Value added by growth; 3. Riskiness of the cash flows; and 4. The time to maturity.	1. Mood and momentum Driven by behavioural factors (panic, fear, greed, etc.) 2. Liquidity and trading ease

	Price of an asset can vary drastically relative to value with variations in liquidity and trading ease. 3. Incremental information Focus is on incremental information (news stories, rumours, gossip, etc.) and how it measures up relative to expectations. 4. Group Think To the extent that pricing is about gauging what other investors will do, price can be determined by the 'herd'.
Investor's Dilemma	Trader's Dilemma
- Uncertainty about the magnitude of the gap - Margin of safety¹⁴ - Collect more information¹⁵ - Ask what if questions - Confront uncertainty - Uncertainty about the closing of the gap - Lengthening your time horizon - Providing or looking for a catalyst that will cause the gap to close	No anchor: You have no inkling of what the value of the stock might be and therefore, no anchor to fall back on to decide whether to hold on or sell a stock. Reactive: Your investment strategy will often be reactive rather than proactive. Crowds are fickle and tough to get a read on: Relative valuation in its essence is the search for shifts in mood of the crowd which is tough.



¹⁴ If you have your margin of safety too high, you are never going to find something to buy.

¹⁵ But remember that the marginal benefits of more information drop off very quickly and may actually end up making you more confused than shed light.

12.5 What is the “Essence of Relative Valuation” (Pricing)?

In relative valuation, the value of an asset is compared to the values assessed by the market for similar or comparable assets.

To do relative valuation then,

- We need to identify comparable assets and obtain market values for these assets;
- Convert these market values into standardized values, since the absolute prices cannot be compared. This process of standardizing creates price multiples;
- Compare the standardized value or multiple for the asset being analyzed to the standardized values for comparable asset, controlling for any differences between firms that might affect the multiple, to judge whether the asset is under- or over-valued.

12.6 What is the allure of relative valuation (why do analysts fall back on relative valuation rather than fundamental valuation)?

There could be three possible reasons that people might prefer pricing over an intrinsic valuation:

1. “If you think I’m crazy, you should see the guy who lives across the hall” – Jerry Seinfeld talking about Kramer in a Seinfeld episode
 - It’s easier to buy and sell something based on relative valuation than a discounted cash flow valuation.
 - Easier to create a story and compare with others than explaining yourself using pure numbers of a discounted cash flow valuation.
2. “A little inaccuracy sometimes saves tons of explanation” – H. H. Munro
 - It’s easier to explain and defend relative valuation than a discounted cash flow valuation.
3. “If you are going to screw up, make sure that you have lots of company” – Ex-portfolio manager
 - In the long term you are going to be more right with an intrinsic valuation than with the relative valuation. But those times you are going to be right alone. And when you are wrong, you are again going to be alone and you will get fired.

12.7 What are the fundamentals of understanding and deconstructing multiples?

Any multiple can be written as:

$$\text{Multiple} = \frac{\text{What you are paying for the asset}}{\text{What you are getting in return}}$$

Below are some of the choices for the numerator and denominator:

Numerator choices	Denominator choices
1. Market value of equity 2. Market value for the firm 3. Market value of the operating assets of the firm (EV)	1. Revenues a. Accounting revenues b. Drivers of value like no. of customers, subscribers, etc. 2. Earnings a. Net income/ EPS b. Operating income 3. Cash flows a. Free cash flow to equity b. Free cash flow to firm 4. Book Value a. Equity b. Firm: BV of Equity + BV of Debt c. Invested Capital: BV of Equity + BV of Debt – Cash

Market Value of Firm = *Market value of Equity + Market value of Debt*

Enterprise Value = *Market value of Equity + Market value of Debt – Cash*

12.8 The four steps to deconstructing multiples

Step 1 Define the multiple	Step 2 Describe the multiple	Step 3 Analyse the multiple	Step 4 Apply the multiple
In use, the same multiple can be defined in different ways by different users. When comparing and using multiples, estimated by someone else, it is critical that we understand how the multiples have been estimated.	Too many people who use a multiple have no idea what its cross sectional distribution is. If you do not know what the cross sectional distribution of a multiple is, it is difficult to look at a number and pass judgment on whether it is too high or low.	It is critical that we understand the fundamentals that drive each multiple, and the nature of the relationship between the multiple and each variable.	Defining the comparable universe and controlling for differences is far more difficult in practice than it is in theory.

13 Relative valuation: Defining the multiple

13.1 Definitional tests

We can test whether the multiple is defined properly by looking at the following questions:

Is the multiple consistently defined?	Is the multiple uniformly estimated?
Proposition 1: Both the value (the numerator) and the standardizing variable (the denominator) should be to the same claimholders in the firm. In other words, the value of equity should be divided by equity earnings or equity book value, and firm value should be divided by firm earnings or book value.	- The variables used in defining the multiple should be estimated uniformly across assets in the “comparable” firm list. - If earnings-based multiples are used, the accounting rules to measure the earnings should be applied consistently across assets. The same rule applies with book value-based multiples.

For example, EV/ EBITDA is consistently defined.

13.2 Why do we net out cash from firm value?

Because the interest income from cash is not considered in EBITDA.

13.3 What happens if a firm has cross holdings which are categorised as a) minority holdings and b) majority active interests?

Minority holdings: Reduce the market value of all cross holdings because income from these is not considered in EBITDA (below the line)

Majority active interests: Add the market value of the minority interest (portion that you do not own) to the numerator because EBITDA includes 100% of the EBITDA of the majority active holding.

14 Relative valuation: Describing the multiple

14.1 Definitional tests

For any multiple that you are analysing, try to answer the following questions:

- What is the average and standard deviation for this multiple across the universe (market).
- What is the median for this multiple?
 - The median for this multiple is often a more reliable comparison point.
- How large are the outliers to the distribution, and how do we deal with the outliers?
 - Throwing out the outliers may seem like an obvious solution, but if the outliers all lie on one side of the distribution (they usually are large positive numbers), this can lead to a biased estimate.
- Are there cases where the multiple cannot be estimated? Will ignoring these cases lead to a biased estimates of the multiple? – example Negative PE ratios.
- How has the multiple changed over time?

Note: Draw a histogram of the data that you collect and you will quickly realise that the distribution is not symmetric – Accordingly, characteristics of a normal distribution should be used with a lot of caution. You cannot trust the Average in case of a skewed distribution.

How to decide what is an appropriate multiple for a sector or industry?

If you end up losing half or two-thirds of the companies in the process of computing that multiple because it cannot be computed for those companies, that multiple is NOT A GOOD MULTIPLE for that sector.

15 Relative valuation: Analysing the multiple

15.1 Analytical tests

For any multiple that you are analysing, try to answer the following questions:

- **What are the fundamentals that determine and drive these multiples?**
 - **Proposition 2:** Embedded in every multiple are all the variables that drive every discounted cash flow valuation – growth, risk and cash flow patterns.
- **How do changes in these fundamentals change the multiple?**
 - The relationship between a fundamental (like growth) and a multiple (such as PE) is almost never linear.
 - **Proposition 3:** It is impossible to properly compare firms on a multiple, if we do not know how fundamentals and the multiple move.

15.2 If we have to worry about all of the fundamental variables even in a relative valuation, why use multiples?

One might ask that if we have to worry about all of the fundamental variables even in a relative valuation then why do it in the first place. There are two reasons for this:

- **Building screens:** Helps in looking for mismatches in the fundamentals and how other companies are being valued given those fundamentals.
- **Focus:** It helps in focusing energy where the odds of finding something mispriced are greater.

15.3 A simple analytical device

How to deconstruct a multiple	if Equity Multiple	if Enterprise Value multiple
Start with a basic intrinsic value model	Start with a dividend or FCFE model	Start with a firm or operating asset model
Divide both sides of the value equation by the denominator of the multiple that you are trying to deconstruct	For instance, for Price-to-Book, divide both sides by book value of equity	For instance, for EV-to-Sales, divide both sides by total sales
You should end up with an intrinsic version of your multiple which relates the multiple to	Intrinsic version of Price-to-Book ratio	Intrinsic version of EV-to-Sales ratio

fundamentals that vary across firms		
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16 Relative valuation: Price earnings ratios

16.1 The fundamental PE ratio equation

In terms of fundamentals, a stable growth PE ratio can be expressed as follows:

$$\text{PE ratio} = \frac{P_0}{EPS_0} = \frac{\text{Payout Ratio} * (1 + g_s)}{(r - g_s)}$$

Payout ratio = 1 – Reinvestment Rate

Accordingly, the fundamental drivers of a PE ratio and their relation to the multiple are:

Variable	Relation
Return on Equity ¹⁶	Positive
Growth (C) ¹⁷	Positive
Beta ¹⁸	Negative

For a high growth firm, the PE ratio can be expressed as follows:

$$\text{PE ratio} = \frac{P_0}{EPS_0} = \frac{\text{Payout Ratio} * (1 + g_h) * \left[1 - \frac{(1 + g_h)^n}{(1 + r)^n}\right]}{(r - g_h)} + \frac{\text{Payout Ratio} * (1 + g_h)^n * (1 + g_s)}{(r - g_s) * (1 + r)^n}$$

16.2 A simple example

Calculating the PE ratios under various scenarios with the following assumptions:

Variable	High growth phase	Stable growth phase
Expected growth rate	25%	8%
Payout ratio	20%	50%
Number of years	5	Forever
Risk free rate	6%	6%

¹⁶ Because payout/ cash flows are derived from return on equity and ROE is a more generic variable.

¹⁷ Represents the comparison variable.

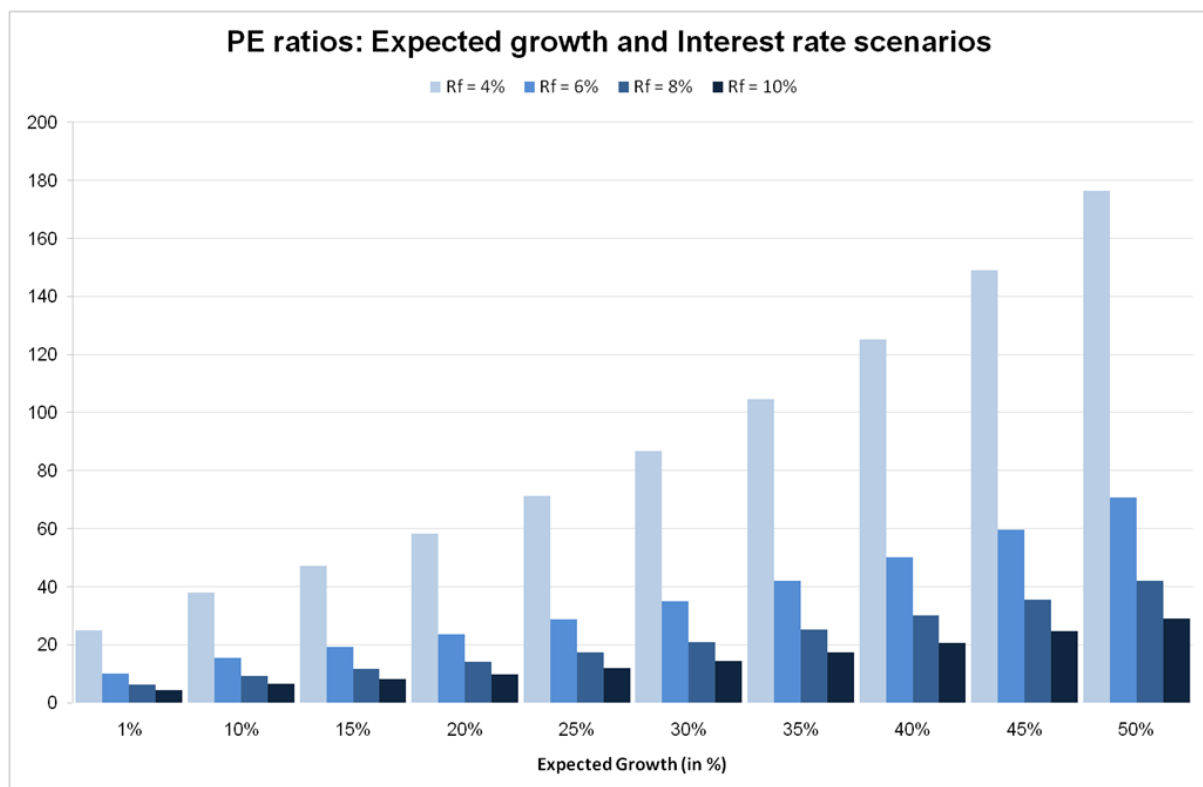
¹⁸ Proxy for risk (i.e., cost of equity/ capital)

Beta	1	1
Equity risk premium	5.50%	5.50%
Cost of equity	11.50%	11.50%

PE ratio based on the above inputs is 28.75. However, we can gauge the sensitivity of the PE ratio to an input by changing values for that input while keeping others constant.

16.3 Relations between PE ratios and various levels of growth and interest rates

Accordingly, we get the following relation between PE ratios between various levels of growth and interest rates:



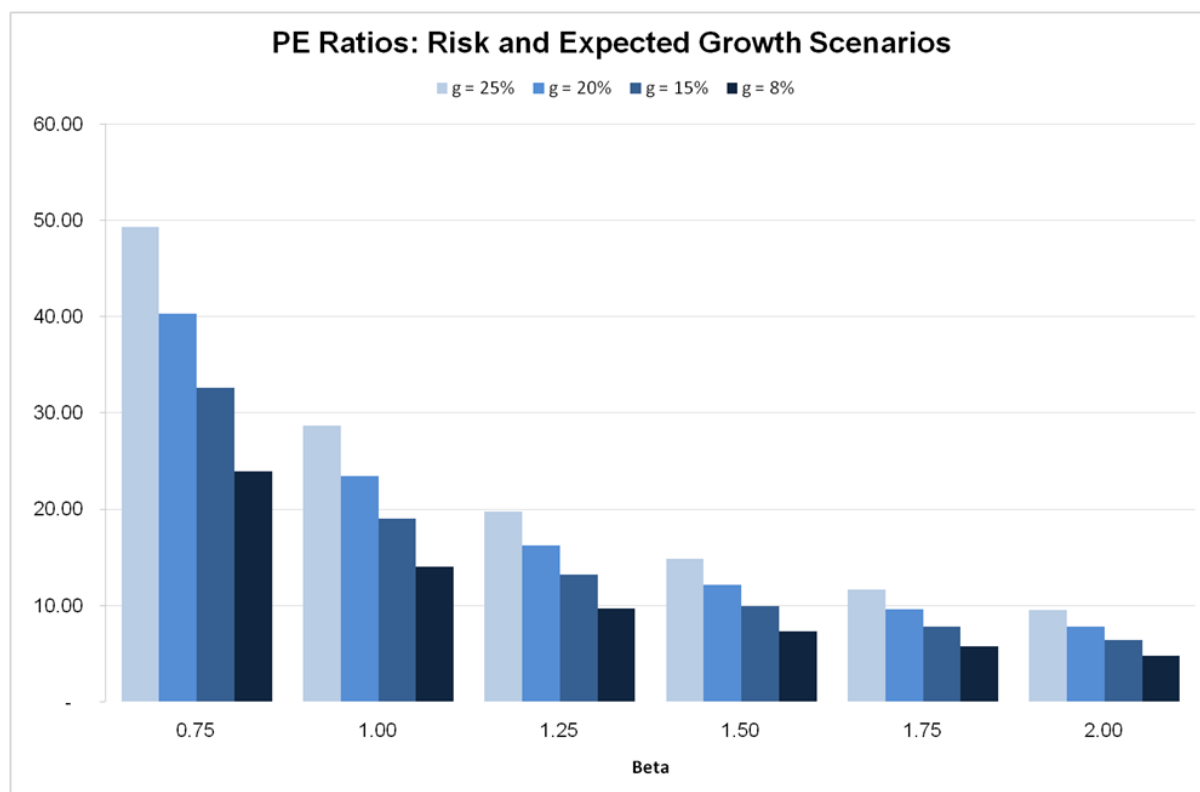
16.3.1 What are the implications/ propositions?

Proposition 1: Under every interest rate scenario, the PE ratio drops with lower growth.

Proposition 2: PE ratio is most sensitive in a LOW interest rate environment. The reason is that the value of growth constitutes a lower proportion of value in a high interest rate environment. As interest rates get lower, across the board growth becomes a bigger proportion of the value of companies.

Therefore, the effect of earnings surprises on markets is more volatile in a low interest rate environment.

16.4 Relations between PE ratios and various levels of growth and risk (represented by Beta)



16.4.1 What are the implications/ propositions?

Proposition 1: Sometimes, the payoff to lowering risk is greater than going for higher growth (example, Amazon in the early 2000s).

16.5 The perfect undervalued company with PE ratios

The perfect undervalued company with PE ratios	
Low PE ratio	High return on equity
High growth	Low beta

16.6 Can you simply conclude that the market is overvalued by just comparing the current PE ratio to its historical average?

NO.

There could be many reasons for the PE ratio to be justifiably high relative to historical average. These reasons have to do with the fundamental factors that affect a PE ratio – growth, cash flows and risk. For instance, the risk free affects the PE ratio of stocks. If risk free rates are low, the expected return that I need to make on stocks is also going to be much lower and vice-versa.

17 Relative valuation: Price to Earnings to Growth ratios (PEG)

17.1 What are the challenges of using a PEG ratio?

A PEG ratio is a PE ratio divided by growth in order to control for growth. However, there are two problems with the PEG ratio:

- PEG ratio assumes a linear relation between PE and growth which is not true; and
- PEG ratio does not control for risk and you may end up with highly risky stocks if you buy stocks based on a low PEG ratio basis.

17.2 PEG ratios and fundamentals: Propositions

Proposition 1: High risk companies will trade at much lower PEG ratios than lower risk companies with the same expected growth rate.

- The company that looks most undervalued on a PEG ratio basis in a sector may be the riskiest firm in the sector.

Proposition 2: Companies that can attain growth more efficiently by investing less in better return projects will have higher PEG ratios than companies that grow at the same rate less efficiently.

- Companies that look cheap on a PEG ratio basis may be companies with high reinvestment rates and poor project returns.

Proposition 3: Companies with very low or very high growth rates will tend to have higher PEG ratios than firms with average growth rates. This bias is worse for low growth stocks.

- PEG ratios do not neutralize the growth effect.

18 Relative valuation: Price to Book ratios

18.1 The fundamental PB ratio equation

In terms of fundamentals, a stable growth PB ratio can be expressed as follows:

$$\text{PB ratio} = \frac{P_0}{BV_0} = \frac{RoE * \text{Payout Ratio}_1}{(r - g_s)}$$

However, we know that $g = (1 - \text{Payout ratio}) * RoE$, accordingly, if we substitute Payout ratio in previous equation with $1 - g/RoE$, we get:

$$\text{PB ratio} = \frac{P_0}{BV_0} = \frac{RoE - g_s}{(r - g_s)}$$

Accordingly, the fundamental drivers of a Price to Book ratio and their relation are:

Variable	Relation
Return on equity (C)	Positive
Growth	Positive
Risk	Negative

19 Relative valuation: Enterprise Value to Invested Capital ratio

19.1 The fundamental EV/ Invested Capital equation

In terms of fundamentals, a stable growth PE ratio can be expressed as follows:

$$\frac{EV}{\text{Invested Capital}} = \frac{EV_0}{BV_0} = \frac{NOPAT * (1 - RIR)}{BV * (r - g_s)}$$

$$\frac{EV}{\text{Invested Capital}} = \frac{EV_0}{BV_0} = \frac{RoC * (1 - RIR)}{(r - g_s)}$$

However, we know that $g = RIR * RoC$, accordingly, if we substitute RIR in previous equation with $1 - g / RoC$, we get:

$$\frac{EV}{\text{Invested Capital}} = \frac{EV_0}{BV_0} = \frac{(RoC - g_s)}{(r - g_s)}$$

Accordingly, the fundamental drivers of a EV to Invested Capital ratio and their relation are:

Variable	Relation
Return on capital (C)	Positive
Growth	Positive
Risk	Negative

20 Relative valuation: EV to EBITDA ratio

20.1 The fundamental EV/ EBITDA equation

In terms of fundamentals, value of operating assets can be written as:

$$EV_0 = \frac{FCFF_1}{(k_c - g)}$$

Rewriting the free cash flow to the firm in terms of EBITDA, the equation can be written as:

$$EV_0 = \frac{EBITDA * (1 - t) + Dep * t - Capex - change in WC}{(k_c - g)}$$

Dividing both sides by EBITDA, we the EV / EBITDA ratio as:

$$\frac{EV}{EBITDA} = \frac{(1 - t)}{(k_c - g)} + \frac{Dep * t}{EBITDA * (k_c - g)} - \frac{(Capex)}{EBITDA * (k_c - g)} - \frac{(Chnage in WC)}{EBITDA * (k_c - g)}$$

Accordingly, the fundamental drivers of a EV / EBITDA ratio are:

Variable	Relation
Cost of capital	Negative
Expected growth rate	Positive
Tax rate (C)	Negative
Reinvestment rate	Negative

20.2 The perfect undervalued company with EV/ EBITDA ratio

The perfect undervalued company with EV / EBITDA	
Low EV / EBITDA	Low tax rate
Low reinvestment	Low risk

21 Relative valuation: EV to Sales ratios

21.1 The fundamental EV/ EBITDA equation

We know that $FCFF = EBIT \cdot (1 - t) \cdot (1 - \text{Reinvestment Rate})$; and $EBIT = \text{Sales} \cdot \text{After tax operating margin}$

$$EV = \frac{FCFF_1}{r - g}$$

Dividing by Sales on both sides, we get:

$$\frac{EV}{Sales} = \frac{\text{Aftertax operating margin} \cdot (1 - RIR)}{(r - g)}$$

For a high growth firm, the equation can be written as:

$$\frac{EV}{Sales} = \text{Aftertax operating margin} \cdot \left[\frac{[(1 - RIR_h) \cdot (1 + g_h) \cdot \left(1 - \frac{(1 + g_h)^n}{(1 + r)^n}\right)]}{r - g_h} + \frac{[(1 - RIR_s) \cdot (1 + g_h)^n \cdot (1 + g_s)]}{(r - g_s) \cdot (1 + r)^n} \right]$$

Accordingly, the fundamental drivers of a PE ratio and their relation to the multiple are:

Variable	Relation
Cost of capital	Negative
Expected growth rate	Positive
Reinvestment rate	Negative
After-tax operating margin (C)	Positive

22 Relative valuation: Applying the multiple

22.1 Which multiple should you use in relative valuation?

Once you have analysed a multiple, the decision then becomes which multiple to choose to do a relative valuation. You should NOT USE a multiple which makes you lose about half of the companies in the sector.

Red Flag

When you find most cheap stocks belonging to one sector, may be something happened in the sector which the market is factoring in but is not getting reflected in your denominator yet (i.e. financials).

22.2 What about disagreement between intrinsic and relative valuation

Suppose a stock is trading at \$15 per share and you come up with a DCF value of \$10 and a relative value of \$20. What would you do? What would your conclusion be?

- The stock is undervalued (its price < relative value)
- The stock is overvalued (its price > intrinsic value)
- The stock is correctly valued (its price = average of intrinsic and relative values)
- The stock is both undervalued and overvalued
- None of the above

The one thing that you should not do is try to be both a pricer as well as a valuer.

22.3 What about disagreement between different relative valuations

Given the above example, now suppose that you run a market regression and come up with a relative of \$10 per share as against a value of \$20 per share based on the peer group of the company. What would your conclusion be?

- The stock is undervalued (its price < peer group relative value)
- The stock is overvalued (its price > market relative value)
- The stock is correctly valued (its price = average of peer and market relative values)
- The stock is both undervalued and overvalued
- None of the above

The conclusion will depend on who you compare yourself to. The conclusion is simply that your stock is undervalued relative to one group of stocks while overvalued relative to another (different) group of stocks. (Think golf against amateurs and professionals).

22.4 Closing thoughts on relative valuation

Proposition 1	Proposition 2	Bottom Line
In a relative valuation, all that you are concluding is that a stock is under or over-valued, RELATIVE TO your comparable group	In asset valuation, there are no similar assets. Every asset is UNIQUE .	Relative valuation is PRICING , not valuation.
Your relative value judgment can be right and your stock hopelessly over valued at the same time.	If you do not control for fundamental differences in risk, cash flows and growth across firms when comparing how they are priced, your valuation conclusions will reflect your flawed judgments rather than market mis-valuations.	

Real Options

23 Real Options (Introduction)

23.1 What are real options?

Real options are investments that are unviable today, however, they have the possibility of becoming viable investments in the future. For instance, think of the following situation:

A small start-up company has come up with a new cell phone technology that promises to speed up download speeds and create smoother interface. The cost of implementing the technology is expected to be \$2 billion and the present value of the expected cash flows from it is only \$1.5 billion. Should they implement the technology now? The answer to this question is NO.

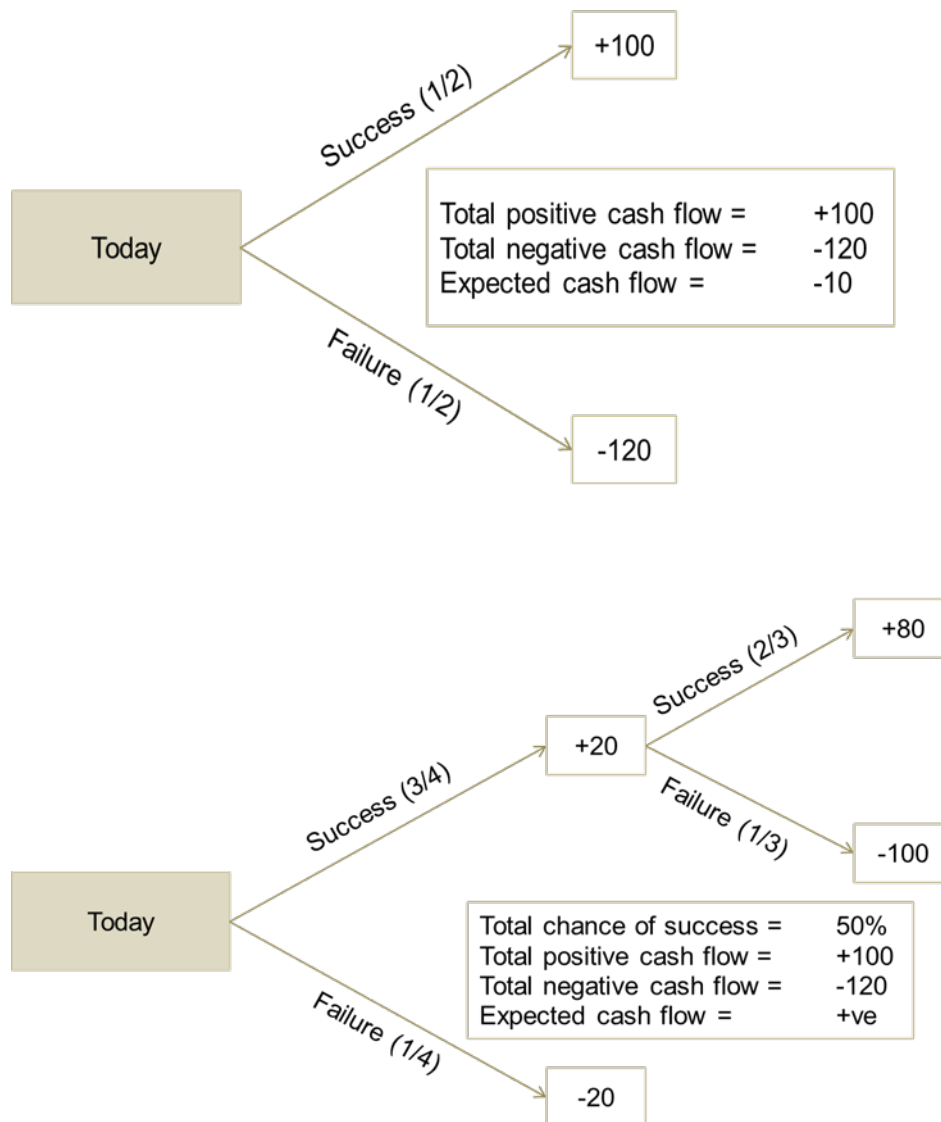
However, what if you are told that you work for an established cell phone manufacturer. Would you be willing to pay for this currently non-viable technology, assuming that the company has patented it and has many years left on the patent life? The answer to this question is **MAYBE. This is the value of the option.**

23.2 What is the underlying theme in real options valuation?

Traditional discounted cash flow models underestimate the value of investments, where there are options embedded in the investments to:

Delay	Flexibility	Expansion	Abandonment
Delay or defer making the investment	Adjust or alter production schedules as price changes	Expand into new markets or products at later stages in the process, based upon observing favourable outcomes at the early stages	Stop production or abandon investments if outcomes are unfavourable at early stages

In essence, real option advocates believe that you should be paying a premium on discounted cash flow value estimates.



23.3 What are the conditions for real options to exist?

Is there a real option EMBEDDED in a decision or an asset?	Does that real option have SIGNIFICANT ECONOMIC VALUE ?	Can that value be ESTIMATED using an option pricing model?
- There has to be a clearly defined UNDERLYING asset - The payoffs on this asset have to be CONTINGENT on a specified event occurring within a FINITE PERIOD	- For an option to have significant economic value, there has to be restriction on competition (exclusivity) in the event of that contingency. - At the limit, real options are most valuable when you have EXCLUSIVITY – you and only you can take advantage of the contingency.	Variables related to the underlying: - Value of the underlying; - Variance in that value; and - Expected dividends on the asset. Variables relating to the option: - Strike price of the option; and - Life of the option

		Macro variable: Level of interest rates¹⁹
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Please refer Appendix A for a primer on option pricing models.

24 Real Options: Options in Projects, Investments and Acquisitions

24.1 What are some of the options that relate to projects, investments and acquisitions?

One of the limitations of traditional investment analysis is that it is static and does not do a good job of capturing the options embedded in investments. There are three types of options that can exist in investment projects:

Option to delay	Option of other opportunities	Option to abandon
Delaying an investment when the firm has exclusive rights to it, until a later date.	Taking one investment may allow us to take advantage of other opportunities (investments) in the future.	Abandon an investment if the cash flows do not match up.

These options all add value to projects and may make a “bad” investment (from traditional analysis) into a good one.

24.2 What is the option to delay?

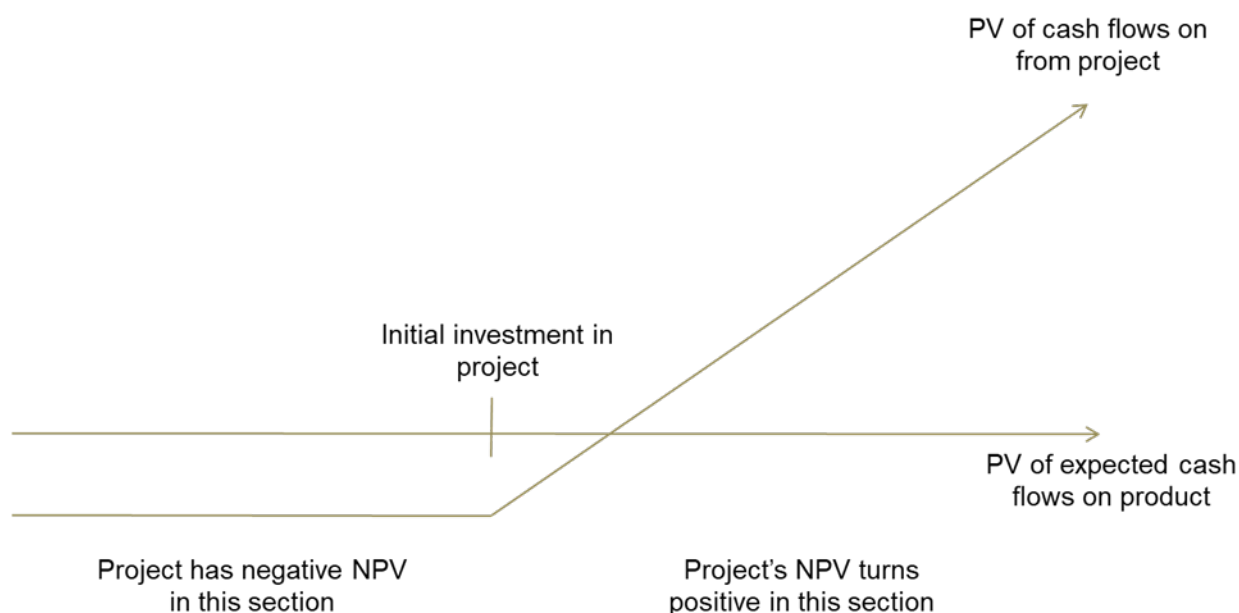
The option to delay is the exclusive RIGHT that an organisation has to undertake a project or develop a product for a specified period of time. This right, then gives the firm the ability to delay taking the project or product until a later date. When a firm has exclusive rights to a project or product for a specified period of time, it can delay taking this project or product until a later date.

A traditional investment analysis just answers the question of whether the project is a “good” one if taken TODAY.

Thus, the fact that a project does not pass muster today, does not mean that the rights to this project are not valuable.

¹⁹ Holding all else constant, as rates increase, the right to buy (sell) at a fixed price in the future becomes more (less) valuable. The reason is that you can invest the cash (that you don't have to pay now) and earn a higher rate in a 30% interest rate environment than in a 3% interest rate environment and still end up paying \$500,000 a year from now. That is, the present value of the strike price is lower in a higher interest rate environment.

Payoff Diagram: Option to Delay



Obtaining inputs for a Patent Valuation

Input	Estimation Process
Value of underlying asset	- Present value of cash inflows from taking the project now. - This will be noisy; but that adds value.
Variance in value of underlying asset	- Variance in cash flows of similar assets or firms; or - Variance in present value from capital budgeting simulation.
Exercise price on option	- Option is exercised when investment is made; - Cost of making investment on the project – assumed to be constant in present value \$
Expiration of the option	Life of the patent
Dividend yield	- Cost of delay; - Each year of delay translates into one year of value-creating cash flows: Annual cost of delay = $1 / n$

24.2.1 Example: Avonex

- Biogen, a bio-technology firm, has a patent on Avoex, a drug to treat multiple sclerosis, for the next 17 years, and it plans to produce and sell the drug by itself.
- Key inputs on the drug are as follows:

Input		
PV of cash flows from introducing the drug now	S	\$3,422 million

PV of cost of developing drug for commercial use	K	\$2,875 million
Patent life	t	17 years
Variance in expected present values	σ^2 (industry average)	0.224
Risk free rate	r = 17-year T-Bond rate	6.7%
Expected cost of delay	y = 1 / 17	5.89%
Output from option pricing model		
d1		1.1362
N(d1)		0.8720
d2		-0.8512
N(d2)		0.2076
Call value	$(3422) * (e^{-0.0589 * 17}) * (0.8720)$ – $(2875) * (e^{-0.067 * 17}) * (0.2076)$	\$907 million

As per this model, Biogen would have been better off to not develop the drug for the next 8 years if everything remained the same. However, it ended up developing the drug 2 years into the process. Why?

Because you are implicitly assuming that for the next 17 years, Biogen is going to be the only game in town to treat this disease. The only thing that the patent does is that it offers you protection against competition only for your own formula. It does not preclude other pharmaceutical companies from doing their own research and come up with a new formula/ drug to treat this disease. Accordingly, the cost of delay should ideally be 1/5 – the estimated time until another drug for the same disease comes in the market – and not 1/17.

24.2.2 How can options pricing models be used in the context of firms that have patents?

The value of a firm with a substantial number of patents can be derived using the option pricing model:

Value of firm		Method used
=	Value of existing products	DCF Valuation
+	Value of existing patents	Option pricing model
+	Value of new patents that will be obtained in the future	Option pricing model

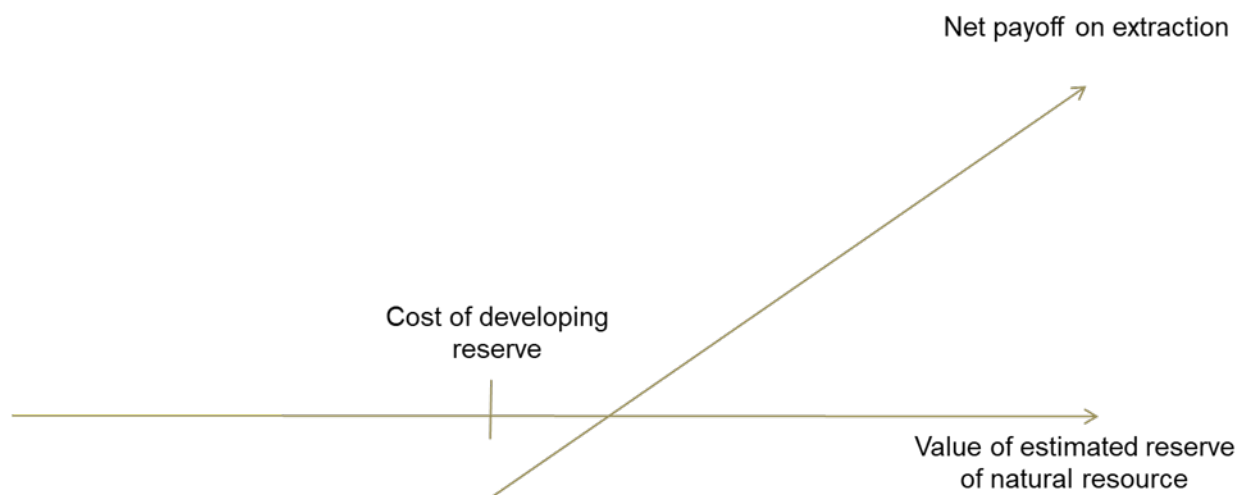
	(-)	
	Cost of obtaining new patents	

Below are some notes to keep in mind:

- The last input measures the efficiency of the firm in converting its R&D into commercial products. If we assume that a firm earns its cost of capital from research, this item will be 0.
- If we use this approach, we should be **careful not to double count** and allow for a higher growth rate in cash flows (in the DCF portion of the valuation).

24.2.3 Options pricing models in the context of natural resource firms

Payoff Diagram: Natural Resource Firms



Obtaining inputs for a Natural Resource option	
Input	Estimation Process
Value of available reserves of the resource (Value of underlying)	• Expert estimates (geologists for oil); • The present value of the after-tax cash flows from the resource are then estimated.
Cost of developing the reserve (Strike price)	• Past costs and the specifics of the investment.
Expiration of the option	• Relinquishment period, if the asset has to be relinquished at a point in time; or • Time to exhaust inventory, based upon inventory and capacity output.
Variance in the value of the underlying asset	• Based upon variability of the price of the resources and variability of available reserves.

Net production revenue (Dividend yield)	Net production revenue every year as percent of market value
Development lag	Calculate the present value of reserve based upon the lag,

Example: Gulf Oil

Gulf oil was the target of a takeover in early 1984 at \$70 per share (it had 165.30 million shares outstanding, and total debt of \$9.9 billion)

Key inputs are as follows:

Input		
Price of oil		\$22.38 per barrel
Production costs, taxes and royalties		\$7 per barrel
Estimated reserves		3,038 million barrels
Average cost of developing the reserves	in present value terms (with development lag of 2 years)	\$10 per barrel
Value of estimated reserves discounted back for period of development lag	$S = 3038 \times (22.38 - 7) / (1.05^2)$	\$42,380.44 million
PV of developing cost of reserves	$K = 3038 \times 10$	\$30,380.00 million
Average relinquishment period	t	12 years
Variance in oil prices	σ^2	0.03
Risk free rate	r	9.00%
Net production revenues	y = % of value of developed reserves	5.00%
Output from option pricing model		
d1		1.6548
N(d1)		0.9510
d2		1.0548
N(d2)		0.8542

Call value (Value of undeveloped reserves)	$(42380.44) * (e^{-0.05 * 12}) * (0.9510)$ $-$ $(30380) * (e^{-0.09 * 12}) * (0.8542)$	\$13,306 million
Traditional DCF value today	42380 – 30380	\$12,000 million

The value of a natural resource firm can therefore be estimated as follows:

Value of firm		Method used
=	Value of already developed reserves	DCF Valuation
+	Value of undeveloped reserves	Option pricing model

Real options test for natural resource options

The options test	
Underlying asset	Natural resource
Contingent Payoff	•
The exclusivity test	
Exclusivity	Scarcity of the natural resource (you can't find oil in your own backyard)
The pricing test	
Underlying asset	
Option	
Cost of exercising the option	

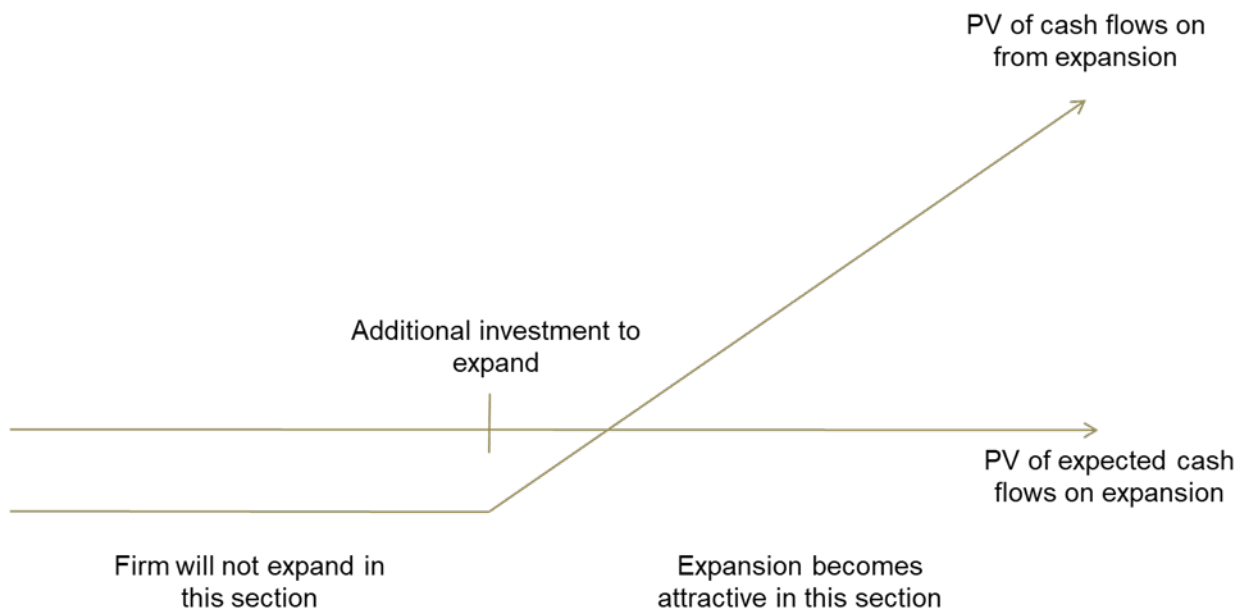
24.3 The option to expand/ take other projects

There are three key aspects to this option:

- Taking a project today may allow a firm to consider and take other valuable projects in the future.
- This, even though a project may have negative NPV, it may be a project worth taking if the option it provides the firm (to take other projects in the future) provides a more-than-compensating value.

- These are the options that firms often call “strategic options” and use a rationale for taking on “negative NPV” or even “negative return” projects.

Payoff Diagram: Option to Expand



24.3.1 An example of the option to expand/ take other projects

Suppose we are valuing a young, start-up company that makes anti-virus software - SecuremMail. You have completed a DCF valuation of this company and estimated a value of \$115 million.

Assume that there is the possibility that the company could use the customer base that it develops for the anti-virus software and the technology on which the software is based to create a database software program sometime in the next 5 years. Key inputs are as follows:

Inputs		
After-tax cash flows from database program	each year for 10 years	\$40 million
Cost of capital	k_c	12.00%
Value of entering database software market	$S = \text{PV of 40 for 10 years @ 12\%}$	\$226 million
Cost of entering database software market	K	\$500 million
Period during which you have right to enter	t	5 years

Standard deviation in firm value	σ (at publicly traded database companies)	50%
Risk free rate	r	3.00%
Output from option pricing model		
Call value (Value of undeveloped reserves)		\$56 million
DCF value	given	\$115 million
Value of company with option to expand	56 + 115	\$171 million

24.3.2 What are some of the key questions to ask when evaluating an option to expand?

One of the most important aspects of options are that OPPORTUNITIES ARE NOT OPTIONS. Having said that, below are some of the key questions to think about when evaluating an option to expand:

Option has no value	Option has value
Is the first investment necessary for the second investment?	
Not necessary	Pre-requisite
Do you have any exclusivity/ competitive advantage on the second investment?	
Zero competitive advantage on second investment	An exclusive right to second investment
Does the second have excess returns?	
Zero excess returns	Large sustainable excess returns

Opportunity + Something exclusive that gives the optionality which may end up having value. For example, Facebook has exclusivity due to its user base.

24.3.3 Real options test for expansion options

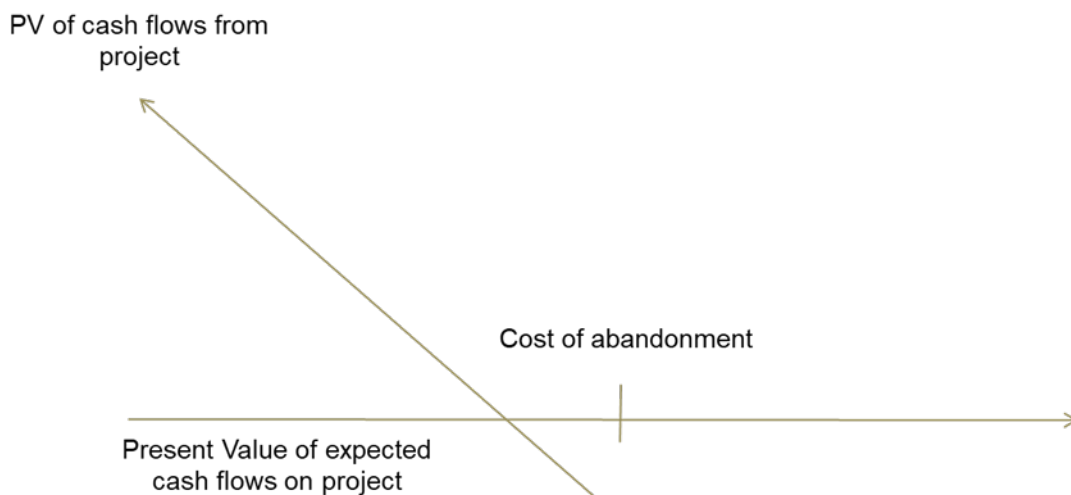
The options test	
Underlying asset	Expansion project
Contingent Payoff	- If PV of CF from expansion > Expansion cost = PV – Expansion cost - If PV of CF from expansion < Expansion cost = 0
The exclusivity test	

Exclusivity	Barriers may range from: • Strong = exclusive licenses granted by the government • Weak = Brand name, knowledge of the market • Weakest = First mover
The pricing test	
Underlying asset	As with patents, there is no trading in the underlying asset and you have to estimate value and volatility
Option	Licenses are sometimes bought and sold, but more expansion options are not
Cost of exercising the option	Not known with any precision and may itself evolve over time as the market evolves.

24.4 The option to abandon

- A firm may sometimes have the option to abandon a project if the cash flows do not measure up to expectations.
- If abandoning the project allows the firm to save itself from further losses, this option can make a project more valuable.

Payoff Diagram: Option to Abandon



24.4.1 An example of the option to abandon: Lear Aircraft and Airbus

- Airbus is considering a joint venture with Lear Aircraft to produce a small commercial airplane (capable of carrying 40-50 passengers on short haul flights).
 - Airbus will have to invest \$500 million for a 50% share of the venture;
 - Its share of the present value of expected cash flows is \$480 million today.

- Lear Aircraft is eager to enter this deal and offers to buy Airbus' 50% share of the investment anytime over the next 5 years for \$400 million, if Airbus decides to get out of the venture.
- A simulation of the cash flows on this time share investment yields a variance in the present value of the cash flows from being in the partnership is 0.16.
- Life of the project is 30 years.

Input (this is a put option)		
Value of the underlying	S	\$480 million
Salvage value from abandonment	K	\$400 million
Time to expiration	t	5 years
Variance in underlying	σ^2	0.16
Risk free rate	r	6.00%
Dividend yield	$y = 1/\text{Life of project} = (1/30)$	0.033
Output from option pricing model		
Put value		\$73.23 million
+ NPV	480 – 500	-\$20 million
NPV with abandonment option		\$53.23 million

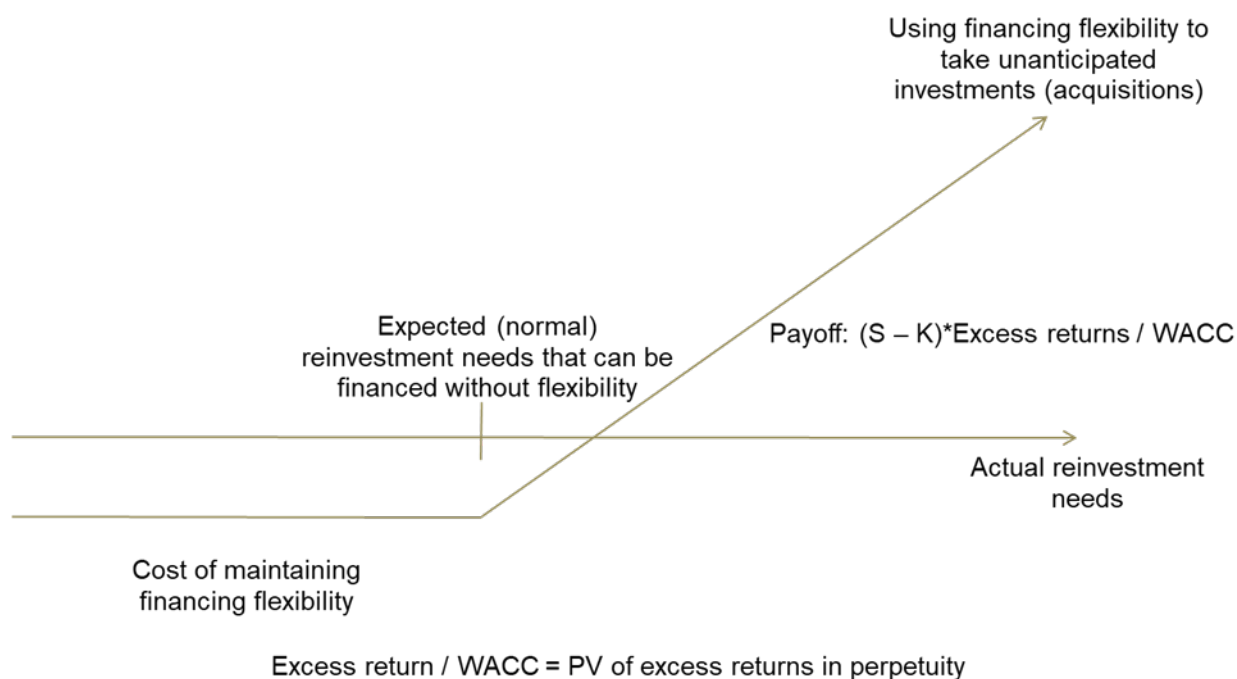
25 Real Options: Options in the capital structure

25.1 How do options occur in the capital structure of a company?

Options in the capital structure can also be thought of as the value of maintaining FLEXIBILITY.

- Firms maintain excess debt capacity or larger cash balances than are warranted by current needs, to meet unexpected future requirements.
- While maintaining this financing flexibility has value to firms, it also has a cost – the excess debt capacity implies that the firm is giving up some value and has a higher cost of capital.
- The value of flexibility can be analysed using the option pricing framework – a firm maintains large cash balances and excess debt capacity in order to have the option to take projects that may arise in the future.

Payoff Diagram: Value of Flexibility



Obtaining inputs for a Flexibility option	
Input	Estimation Process
Value of underlying (S)	- Expected annual reinvestment needs (as a % of firm value). - Measures magnitude of reinvestment needs.
Strike Price (K)	(Internal + Normal access to external funds) / Value - Measures the capital constraint
Expiration of the option	1 year - Measures an annual value for flexibility
Variance	- Variance in annual reinvestment needs. - Measures how much volatility there is in investment needs.

25.2 An example of options in the capital structure: Disney's financial flexibility

Below are some of the inputs for Disney:

- Disney's optimal debt ratio = 40%; Cost of capital = 11.64%
- Disney's actual debt ratio = 18%; Cost of capital = 12.22%
- Annualised cost of flexibility = 12.22% - 11.64% = 0.58%

Inputs		
S <i>Expected annual reinvestment needs (as a % of firm value)</i>	Average (Reinvestment / Value) over last 5 years	5.30%

K <i>(Internal + Normal access to external funds) / Value</i>	Average over last 5 years	4.80%
Expiration of the option		1 year
Variance	Variance over last 5 years in $\ln(\text{Reinvestment} / \text{Value})$	0.375
Output		
Value of option	Value of the option to take the project <i>but over value of flexibility will depend on the quality of the projects taken</i>	1.6092%
Historical returns on projects		18.69%
Cost of capital		12.22%
Excess returns	18.69% - 12.22%	6.47%
Value of flexibility (annual)	$1.6092\% \times (0.0647 / 0.1222)$	0.85%
Decision	Compare 0.85% vs. 0.58%	Maintain flexibility

25.3 What are the determinants of the value of flexibility?

1. **Capital Constraints (external and internal):** The greater the capacity to raise funds, either internally or externally, the less the value of flexibility.
 - Firms with significant internal operating cash flows should have flexibility less than firms with small or negative operating cash flows; and
 - Firms with easy access to financial markets should have lower value for flexibility than firms without that access – private companies will value flexibility more than public companies.
2. **Unpredictability of reinvestment needs:** The more unpredictable the reinvestment needs of a firm, the greater the value of flexibility.
3. **Capacity to earn excess returns:** The greater the capacity to earn excess returns, the greater the value of flexibility.
 - Firms that do not have the capacity to earn or sustain excess returns get **no value from flexibility**.

26 Real Options: Equity as an option

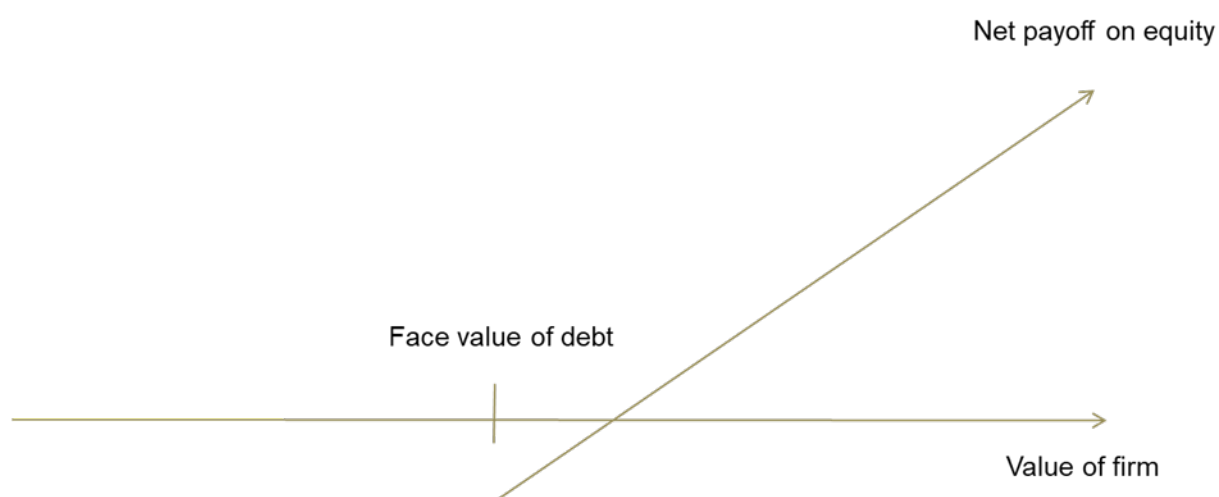
26.1 Why can equity in a business be viewed as an option?

Equity in a firm is a residual claim, i.e., equity holders lay claim to all cash flows left over after other financial claimholders (debt, preferred stock, etc.) have been satisfied.

If a firm is liquidated, the same principle applies, with equity investors receiving whatever is left over in the firm after all outstanding debts and other financial claims are paid off.

The **principle of limited liability**, however, protects equity investors in publicly traded firms if the value of the firm is less than the value of the outstanding debt, and they cannot lose more than their investment in the firm.

Payoff Diagram: Option to Liquidate



26.2 Example of equity as an option

- Assume that you have a firm whose assets are currently valued at \$100 million and that the standard deviation in this asset value is 40%.
- Further, assume that the face value of debt is \$80 million (it is a zero coupon bond with 10 years left to maturity).
- If the ten year treasury bond rate is 10%,
 - How much is the equity worth? And
 - What should the interest rate on debt be?

Obtaining inputs for a Liquidation option		
Input	Estimation Process	Value

Value of underlying (S)	Value of firm	\$100 million
Strike Price (K)	Face value of outstanding debt	\$80 million
Expiration of the option	Life of zero coupon debt	10 years
Variance	Variance in firm value	0.16
Risk free rate	Treasury bond rate corresponding to option life	10%
Output from the model		
d1		1.5994
N(d1)		0.9451
d2		0.3345
N(d2)		0.6310
Value of call (i.e. value of equity)	$100 \times 0.9451 - 80 \times e^{-0.10 \times 10} \times 0.6310$	\$75.94 million
Value of outstanding debt	$100 - 75.94$	\$24.06 million
Interest rate on debt	$(80 / 24.06)^{1/10} - 1$	12.77%
Probability of default	$1 - N(d2)$	36.90%

26.3 What are some of the effects of a catastrophic drop in value?

- The option characteristics of equity enable equity to have some value and the debt holders share some of the loss in value.
- The two things which allow equity to have value are – HOPE and TIME.

26.4 Inputs for equity as an option

Obtaining inputs for a Liquidation option	
Input	Estimation Process
Value of underlying (Value of firm)	• Cumulate market values of equity and debt; or • Value the assets in place using FCFF and WACC; or

	Use cumulated market values of assets, if traded.
Value of debt	- If debt is short term, you can use only the face value or book value of the debt; - If the debt is long term and coupon bearing, add the cumulated nominal value of these coupons to the face value of the debt.
Maturity of debt	- Face value weighted duration of bonds outstanding; or - If not available, use weighted maturity.
Variance	- If stocks and bonds are traded: $\sigma_{\text{firm}}^2 = w_e^2 \sigma_e^2 + w_d^2 \sigma_d^2 - 2 \rho_{ed} w_e w_d \sigma_e \sigma_d$ - If not traded, use variances of similarly rated bonds. - Use average firm values variance from the industry in which company operates.

26.5 What is N(d1) and N(d2)? What is the probability that S>K?

It is N(d2) which is the risk neutral probability that S>K. In some of the problems, though, we have used a range from N(d1) and N(d2) as the range of probabilities. And the reason is this: N(d1), in addition to being an option delta, is also a probability that the option will be in the money. **In fact, the only reason d1 is different from d2 is because you are uncertain about S.**

$$d2 = d1 - \sigma\sqrt{t}$$

If you had a standard deviation of zero, N(d1) = N(d2). As the uncertainty increases, the gap between these two numbers will widen. **Thus, you go from being certain about the probability to having a range. Having said all of this, N(d2) should be the point estimate on the probability that S>K. You can use the range to indicate that there is uncertainty about this probability.**

26.6 What is the cost of delay?

This is a tough one. Sometimes, we use 1/n and sometimes we use the cash flow next year / S and sometimes we use no cost of delay at all. Let's look at the conceptual basis. The cost of delay is a measure of how much you will lose in the next period if you don't exercise the option now as a fraction of the current value of the underlying asset (it parallels the dividend yield. On a listed option on a stock, if you exercise, you will have the stock and get the dividends in the next period). Thus, if you have a viable oil reserve, the cost of delay is the cash flow you would have made on the developed reserve next period divided by the value of the reserve today.

Here is the overall rule you should adopt. If you have a decent estimate of the cashflows you will receive each period from exercising the option, it is better to use that cashflow/ PV of the asset as the dividend yield. If your cash flows are uneven or if you do not know what the cashflows will be each period, you should use 1/n as your cost of delay. If you lose nothing in terms of cashflows by waiting, you should have no cost of delay.

Let me take three examples. The first is the bidding rights to televise the Olympics in an earlier quiz. There were two years left to the Olympics and you were trying to price the option. In this case, there is no cost of delay since you really cannot exercise the option early even if it is deep in the money – you cannot televise the Olympics a year before they happen. The second is the oil reserve option. Since the cashflows from the reserve tend to be fairly uniform over time (based upon the barrels of oil you would produce and the current price per barrel, it is easy to estimate the cashflows you would generate each year on the reserve. In most of the oil reserve problems, therefore, you would go with the cashflow/ PV of oil in the reserve as your cost of delay. The third is the patent examples. While you may be able to estimate the expected cashflow each year from commercialising the patent, these cashflows are more difficult to obtain and are less likely to be uniform over time. That is why many of the patent problems use the less preferred option of $1/n$ as the cost of delay, where n is the number of years left in the patent.

Early exercise and the cost of delay: With listed options, you are almost always better off holding or selling an option than exercising it. That is why most options can be valued through their stated maturities. With real options, though, not exercising until expiration sometimes makes no sense, since the value of the real option comes from exercise. Thus, if you have the right to develop an oil reserve for the next 20 years, it does not make sense to wait until the last day of the twentieth year to develop the reserve, since you will never get the oil out in a day. That is why you see a cost of delay introduced as a counterweight. Intuitively, think of this as the cost of not exercising a real option, once it becomes viable. Thus, the cost of delay with a viable oil reserve is the cash flow that you would get by developing it immediately and if that cash flow exceeds the time premium on the option, you will develop. This cost of delay operates like a dividend on a stock, with options outstanding, and that is why you have to use the dividend-yield adjusted option pricing model.

Ultimately, the real insight you get from real options is not in the option pricing models but in the intuition. They explain why vulture investors buy equity in the most risky distressed sectors and not the least risky, why companies may pay large amounts for non-viable technologies and why pharmaceuticals may hold off on developing viable drugs. They may also explain why you pay premiums for some young technology companies, over and above the discounted cash flow value. Until next time!

27 Value Enhancement

27.1 What does value enhancement mean?

Value enhancement means taking actions which affect the value (and not necessarily the price) of the business by affecting any of the following drivers of value:

- Cash flows
- Risk (i.e., discount rate)
- Growth rate
- Length of high growth period

27.2 What is the difference between price enhancement vs. value enhancement?

To enhance the price of a stock, the managers of the firm must keep the market happy. And what keeps the market happy depends on the current mood of the markets and changes from time to time. Value enhancement, as mentioned in the previous question, means taking actions which affect VALUE of a business, even though it may not get reflected in the PRICE that the market attaches immediately.

27.3 Which of the following actions is a “value-enhancing” decision?

Action	Value enhancing or not
Stock split	No
Amortizing goodwill	No (Yes, if tax deductible)
Changing depreciation methods	No (Maybe depending on whether tax books affected or not)
Issuing tracking stock ²⁰ on your “social media” business	No

However, any of the above decisions could lead to a change in the “PRICE” of a stock.

27.4 If you were the CEO of a multi-business company, which division would you sell if you had to? (some information regarding each of the businesses is given).

Division	Return on Capital	Cost of Capital	Which to sell?
A	20.00%	10.00%	Could be any one of these businesses. Depends on the price that you can get for a particular business versus the value of that business.
B	15.00%	9.00%	
C	8.50%	8.50%	
D	3.00%	7.50%	<i>It is quite possible that people are more dazzled by your best business and end</i>

²⁰ Issuing shares on a segment of the business (e.g., online tracking stocks in late 90s) – generally issued on a current hot sector.

			<i>up paying a higher price relative to the value of that business.</i>
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27.5 How do stock buy-backs affect value of a firm?

Stock buybacks may affect the following two areas of value:

- **Capital Structure:** Stock buybacks lower the equity in the company and thus lead to higher debt in the capital structure. This has a direct impact on the cost of capital of the firm.
- **Companies which have a history of taking bad projects:** May change the perception of investors and stupidity discount of a business.

27.6 What are the paths to value enhancement (how can value of a business be enhanced)?

We can use the DCF framework to determine the ways in which value can be enhanced:

Focus Area	Actions
Cash Flows	• Increasing after-tax operating income • Reducing reinvestment needs of the business
Growth Rate	• Increasing (decreasing) the rate of reinvestment in the firm if the firm earns a Return on Capital which is more (less) than its Cost of Capital • Improving the return on capital of the business
Length of the Growth Period	• Take actions that enhance the competitive advantages of the business which will allow it to earn higher returns for longer
Riskiness (Cost of Capital)	• Reducing the operating risk in investments/ assets (ratio of fixed vs. variable costs) • Changing the financial mix (ratio of debt and equity in the capital structure) • Changing the financing composition

Each of these focus areas have been discussed in further detail below:

27.6.1 Increasing cash flows from assets in place:

Area	What can you do?
Revenues	
* Operating margin	More efficient operations and cost cutting – higher margins
= EBIT	Divest assets that have negative EBIT
– Taxes	Reduce tax rate by: • Moving to lower tax locales • Transfer pricing • Risk management
= $EBIT \cdot (1 - t)$	
– Capex – Depreciation	Live off of past over-investment
– Change in WC	Better inventory management and tighter credit policies

=	Free cash flow to firm	
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27.6.2 Increase expected growth:

Area		What can you do?
	Reinvestment rate	- Reinvest more in projects - Do acquisitions
*	Return on capital	- Increase operating margin - Increase capital-turnover ratio
=	Expected growth rate	

Things to think about:

- How will your competitors react to your moves?
- How will you react to your competitors' moves?

Modes of organic growth vary in value creation intensity: Consumer goods industry

Category of growth	Shareholder Value	Revenue growth required
New-Product/ Market development	1.75 – 2.00	5 – 6
Expanding an existing market	0.30 – 0.75	13 – 33
Maintaining a growing share of a growing market	0.10 – 0.50	20 – 100
Competing for share in a stable market	– 0.25 – 0.40	n/m – 25
Acquisition (25 th to 75 th percentile result)	– 0.50 – 0.20	n/m – 50
Lesson: Not all growth strategies are created alike. They may seem equal but history suggests otherwise.		

27.6.3 Increase the length of the growth period:

The length of the growth period is inextricably linked to the competitive advantages that the firm enjoys. Some competitive advantages that a firm can possess:

Building on competitive advantages		Find new competitive advantages	
Brand name	Legal protection	Switching costs	Cost advantages

27.6.4 Reducing the cost of capital:

Area		What can you do?
	Cost of Equity	- Make product or service less discretionary (change product characteristics; more effective advertising) - Reduce operating leverage (outsourcing; flexible wage contracts)
*	Weight of Equity	Change financing mix (for instance, add more debt)

+	Weight of debt	
*	Cost of debt	• Match debt to assets • Reduce default risk (swaps, derivatives, hybrids)
=	Cost of Capital	

27.7 Why do voting and non-voting shares have different values²¹?

I hope you've had a chance to see Prof.'s announcement regarding the value of control and the voting/ non-voting issue. But I would just like to add to it.

Professor's answer: The value of voting versus non-voting shares: Some of you have been noting and commenting on my approach for estimating the value of voting shares, where I value non-voting shares by dividing the status quo value by the total number of shares and the expected value of control by just the voting shares. That is an extreme perspective on control, where voting shareholders get the entire value of control, usually in the form of a premium in a hostile acquisition by someone who plans to change the way the company is run. So, you should think of it as the maximum value for a voting share. In reality, even non-voting shares get a portion of the value of control, but a smaller portion than the percent of the firm that they own.

The voting shareholders are not ripping off profits from the non-voting shareholders. Let us understand this issue in the following two scenarios:

27.7.1 Company runs itself better to reach optimal value

In this scenario, the value increase due to better management will be shared by all shareholders alike and accordingly, value of both voting and non-voting shares will converge towards \$10.40. This is the scenario you are talking about.

27.7.2 Two-tier takeover offer

In this scenario, a potential acquirer looks at the company and feels that he can run the company better than the existing management. Thus, he is willing to pay a premium for the shares if he can acquire control of the company.

If the laws of the land do not prohibit two-tier offers, the potential acquirer can get away with only **offering a premium to only the voting shareholders**. A practical example of this that Prof. likes to share is a Brazilian company called Ambev. In a takeover offer, its voting shareholders were offered a price which was 50% more than the price offered to the non-voting shareholders. As a non-voting shareholder, you will be helpless (unless the securities laws protect you) if the voting shareholders decide to take the offer.

This is the situation that Prof. is talking about. And this type of transaction is not just restricted to developing countries as there have been examples of such acquisitions in Canada as well.

As mentioned earlier, the true value will lie somewhere in between.

²¹ https://papers.ssrn.com/sol3/papers.cfm?abstract_id=837405

28 Narratives to Numbers: Connecting Stories to Value

28.1 What is an example of connecting stories to value

Suppose you value a company and use the following information/ assumptions:

Dimension	Assumption
Growth rate	50% for the next 10 years
Operating margins	20% (twice the industry average)
Reinvestment needs	Low (given the growth rate)
Cost of capital	7% in year 10 (below the market average)

Based on the above assumptions, the implicit story you are telling about this company is:

Dimension	What's the story		Why
Company Size	Small or large?	Small	High growth rate
Total market	Stable or growing?	Growing	High growth rate
Competitive advantages	Strong or weak?	Strong	High operating margin
Capital intensity of business	Low or high?	Low	Reinvestment is low
Nature of product/ service	Discretionary or non-discretionary?	Non-discretionary	Low cost of capital

Now, if you were to say that this company is Coca Cola, this would be a fairy tale valuation – not possible in this lifetime. Deconstructing the story from your numbers is the most effective way to avoid fairy tale valuations.

28.2 What are some of the factors that can affect the story of a company?

While it is impossible to an exhaustive list of factors that affect the story of a company, below are some examples of these factors:

- Earnings reports from the company
- Macroeconomic changes (interest rates, inflation, etc.)
- Regulatory changes
- Earnings reports from other companies (competitors)

28.3 How are stories and numbers impacted by the lifecycle stage of a company in the context of valuation?

Valuation is a combination of a story with numbers. Think about the following statements:

- The mix of stories and numbers should change as companies age;
- Stories are more important for young companies and numbers for older companies.

The following table summarises the above thoughts:

Story	Company	Age (in human terms)	Numbers
	Snap	A promising 3-year old	
	Facebook	A precocious teenager	
	Google	In the prime: 30 years	
	Apple	A wealthy 45-year old	
	Microsoft	Trying to be young at 56	
	IBM	Old man: 75 years	

28.4 What are some of the steps in building a narrative

Step 1: Survey the landscape

Every valuation starts with a narrative, a story that you see unfolding for your company in the future. In developing this narrative, you will be making assessments of the current situation of the company:

- Your company (its products, its management and its history);
- The market or markets that you see it growing in;
- The competition it faces and will face; and
- The macro environment in which it operates.

Step 2: Create a narrative for the future

When creating a narrative for the company, keep the following rules in mind:

- Rule No. 1: Keep it simple
- Rule No. 2: Keep it focused – show the pathway to profitability
- Rule No. 3: Stay grounded in reality

The Uber Narrative: June 2014	
The company	An urban car service business: Primarily a force in urban areas and only in the car service business
The market	Would expand the business moderately (about 40% over ten years) by bringing in new users

Competition/ Competitive advantages	Local networking benefits: If it becomes large enough in one city, it will become larger quickly, but that will be of little help when it enters a new city. Maintain revenue sharing (20%) system due to strong competitive advantages (from being first mover – capital) Low capital-intensive business model, with drivers as contractors and very little investment in infrastructure.
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Step 3: Check the narrative against history, economic principles and common sense

Probability of Occurrence		
Cannot assess	Low	Increasing
"IT" is Possible	"IT" is Plausible	"IT" is Probable
"It" could happen, but you are not sure what "it" is, when "it" will happen and what "it" will look like when it does. VALUATION RESPONSE Value as an option, with the value increasing with the size of possible market and the exclusivity of your firm's access to that market. Some examples of THE IMPOSSIBLE • Bigger than the economy: Assuming growth rates for the company bigger than the growth rate for the economy in perpetuity • Bigger than the total market: Allowing a company's revenues to grow so much that it has more than a 100% market share of whatever business it is in. • Profit margin > 100%: Assuming earnings growth will exceed revenue growth for a long enough period, and pushing margins above 100%	Something that you can make a reasoned argument could happen, though you have no tangible evidence for it happening (yet) VALUATION RESPONSE Show as expected growth, adjusting for risk in your expected return. Value will increase with size of the market and your firm's competitive advantages. Some examples of THE IMPLAUSIBLE • Growth without reinvestment: Assuming growth forever without reinvestment • Profits without competition: Assuming that your company will grow and earn higher profits, with no competition. • Returns without risk: Assuming that you can generate high returns in a business with no risk.	This is something that you expect to happen, with some basis or evidence for that expectation. There can be substantial uncertainty in your expectations. VALUATION RESPONSE Show in base year numbers and expected cash flows, adjusting for risk in your expected return. Some examples of THE IMPROBABLE • High growth and Low Reinvestment • High Growth and Low Risk • Low Risk and High Reinvestment

Step 4: Value the company

Once you have the narrative nailed down, put it into numbers and come up with a value for the company.

Step 5: Keep the feedback loop open

Be on the lookout for changes to your story and story breakdowns, and adjust your valuation accordingly.

29 Asset-based valuations

29.1 Under what circumstances does asset-based valuations make sense?

There are three situations in which an asset-based valuation would make sense:

Liquidation	Accounting Mission	Sum-of-the-Parts valuation
When you wish to liquidate a business by selling its assets rather than as a composite business	Both US GAAP and IFRS have turned to “fair value” accounting to reflect assets at their fair value rather than book value	Value parts of business individually if business made up of individual divisions or assets. This may be done for two groups: • Potential acquirers – as a precursor to restructuring the business; or • Investors – who may be interested because the business is selling for less than the sum of its parts.

Liquidation valuation

- In liquidation valuation, you are trying to assess how much you would get from selling the assets TODAY, rather than the business as a going concern.
- Consequently, it makes more sense to do a RELATIVE valuation than an intrinsic valuation.
- To the extent that the liquidation is URGENT, you may attach a DISCOUNT to estimate value.

Accounting valuation

- The concept of “market participant”: Fair value is defined as PRICE that a company would receive if it were sold to a third party TODAY. (this is not fair value, rather fair PRICE).
- However, required to also provide a backing for that “price” with a discounted cash flow valuation! – This leads to reverse engineering with intrinsic valuations – VALUE FIRST, VALUATION LATER

Sum of the parts valuations

Sum-of-the-parts valuation if individual assets can be done on:

Intrinsic value basis	Relative value basis
• Long term, passive investor • Interested in finding market mistakes	• Activist investor • Plan to acquire or push for change

29.2 When are asset-based valuations easiest to do?

Asset-based valuations are easiest to do in the following circumstances:

Separable assets	Standalone earnings/ cash flows	Active markets for similar assets
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• Easier to value real estate • Difficult to value brand name	• Easier to value if earnings can be traced • Difficult to value where earnings cannot be easily attributed to individual assets	• Easier to value if you can find market for “similar assets” in case of relative valuation
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Private Company Valuations

30 Valuing Private Businesses (Introduction)

30.1 What are some of the key differences between valuing public and private businesses?

While the fundamentals of valuing private businesses are the same as that of valuing public businesses, there are certain nuances or adjustments that we make when valuing private businesses, specifically to do with:

- Cost of equity for private businesses; and
- Cost of illiquidity since private businesses are not actively traded on a public market.

We discuss both issues in detail in the following sections.

30.1.1 Cost of equity for private businesses

We relax the assumption of a diversified investor in the case of a private business which allows us to think in terms of total risk and not just systematic risk. Accordingly, cost of equity will be higher for a private company as compared to a public company which is in the same business and has the same financial practices.

30.1.2 Cost of illiquidity

We incorporate a cost of illiquidity when valuing private businesses. However, we diverge from the usual practice of applying a constant illiquidity discount (e.g. 20%). The cost of illiquidity should be different across:

Companies	Time	Buyers
Money making/ cash flow generating businesses should worry less about liquidity than unprofitable/ negative cash flow companies.	Liquidity matters more at times of crisis and less in normal times. When everyone wants cash, liquidity matters a lot.	Long term buyers with less cash needs should care less about liquidity than short term buyers with greater cash needs.

30.2 Who could be the best potential buyer of a private business?

Assume that the owner of a private business is trying to sell its business. Which of the following buyers is likely to offer the highest price:

- A private owner;
- A private equity fund;
- A publicly traded company; or
- Its unclear.

The answer depends on the cost of equity that the buyer attaches to the business that is up for sale. A public company is likely to have a lower cost of equity for this business because it cares mostly about systematic risk. The only situation in which a private equity owner would pay a higher price is if it believes it can change the cash flows of the company.

This is one of the few places in valuation when two people looking at the same company can have different valuations even though they agree on the cash flows.

30.3 How are the mechanics of valuing private businesses different from valuing public businesses?

The process of valuing a private company is similar to valuing public companies. However, there are two fundamental issues that you are likely to face when valuing private businesses:

Market values	Historical data
There are no market values of debt and equity.	Financial statements of private companies is likely to: - go back fewer years; - have less detail; and - have more holes in them.

30.4 How is having access to market values useful in valuation?

When valuing companies, we use market values in a number of places:

Market values as inputs:

- Debt ratios for going from unlevered betas and for computing cost of capital.
- Market prices to compute the value of options and warrants granted to employees.

Market values as output:

Market values act as a measure of reasonableness. In private company valuations, the value stands alone.

Market price based risk measures:

Such as betas and bond ratings will not be available for private companies.

30.5 What are some of the cash flow estimation issues in private company valuations?

- Shorter history
- Different accounting standards
- Intermingling of personal and business expenses
- Separating 'salaries' from 'dividends'

30.6 What are some of the motives for valuing a private business?

The motive for valuing a private business matters because it affects how you approach a valuation:

You can value a private company for	You can value a division or divisions of a publicly traded firm
'Show' valuations (no money behind them): • Curiosity • Legal purpose: Estate tax and divorce court Transaction valuations: • Sale or prospective sale to another individual or private entity; • Sale of one partner's interest to another; and • Sale to a publicly traded firm. As a prelude to setting the offering price in an IPO.	• As a prelude to a spin off; • For sale to another entity; and • To do a sum-of-the-parts valuation to determine whether a firm will be worth more broken up or if it is being efficiently run.

30.7 What are the main scenarios for a private business valuation?

- Private to private transactions
- Private to public transactions
- Private to IPO
- Private to VC to public

31 Private to private transactions

31.1 Key issues in a private to private transaction?

In a private to private transaction, a private business is sold by one individual to another. There are three key issues that we need to confront in such situations:

Neither buyer nor the seller is DIVERSIFIED	The investment is ILLIQUID	Value of KEY PERSON
Risk and return models that focus on just systematic risk will seriously underestimate the discount rates.	The buyer of the business will have to factor in an "illiquidity discount" to estimate the value of the business.	There may be significant personal component to value (restaurant owned by renowned chef). In other words, revenues and operating profit of the business reflect not just the potential of the business but also the presence of the current owner.

31.2 Step-by-step valuation process with the help of an example of valuing a restaurant.

The setup:

- An investment banker wants to buy an upscale French restaurant whose chef is also the owner of the restaurant. Both restaurant and chef are well regarded, and the business has been good for last 3 years.
- The investment banker will put all of his savings into this restaurant (i.e. he is undiversified).
- No conventional debt outstanding but it has lease commitment of \$120,000 each year for the next 12 years. The financial statements of this restaurant are given below:

Past income statements (in \$ '000)				
Particulars	Y(-3)	Y(-2)	Y(-1)	Remarks
Revenues	800	1,100	1,200	Operating at full capacity
- Operating expenses	120	120	120	12 years left on the lease
- Wages	180	200	200	Owner does not draw salary
- Material	200	275	300	25% of revenues
- Other operating expenses	120	165	180	15% of revenues
Operating income	180	340	400	
- Taxes	72	136	160	40% tax rate
Net income	108	204	240	

Below we discuss the cleanups that need to be done in this case.

31.2.1 Step 1: Estimating discount rates

The issues:

- As a private business, this restaurant has no market price or returns to use in estimation of beta; and
- The buyer is not diversified.

The solution for beta: Using bottom up betas to get unlevered betas.

- Unlevered betas across 75 publicly traded restaurants in the US is 0.86
- However, most of the publicly traded restaurants are fast food chains or mass restaurants. There could be an argument that the beta for an upscale restaurant is more likely to reflect the business risk of high-end specialty retailers than of fast food restaurants – the unlevered beta for 45 high-end retailers is 1.18.

The solution for diversification: capture total risk and not just systematic risk (total beta)

- Suppose the risk in a business is bifurcated as follows:

Total risk (100) divided into	
80 units from firm-specific risk	20 units from market risk

The investment banker is likely to demand a cost of equity that reflects the total risk and not just the systematic risk since all his capital is tied up in this investment.

- To get to total beta from market beta, we need a measure of how much of the risk comes from market and how much is firm specific.
- Looking at the regressions of publicly traded companies that yield bottom-up beta should provide an answer:
 - The average R-squared across high-end retailers regression is 25%
 - Since betas are based on standard deviation (rather than variances), we will take the **correlation coefficient (the square root of the R-squared) as our measure of the proportion of risk that is market risk**. Square root of 25% is 50%. Therefore, 50% of the risk in such a business is explained by market risk.
- Total unlevered beta = Market beta/ Correlation with market = $1.18 / 0.50 = 2.36$

The solution for market prices: No market prices to compute debt ratios

- We have two alternatives to relever the betas in the absence of market values of debt and equity:
 - Assume that the debt to equity ratio for the firm is similar to the average market debt to equity ratio for publicly traded firms in the sector; or
 - Use your estimates of the value of debt and equity as the weights in the computation. (There will be circular reasoning which can be taken care of by excel). – This is the preferred method.
- We get a beta of 2.56 after adjusting for leverage. And cost of equity = 14.50%
- Estimate a cost of debt based on a synthetic rating which gives us a cost of capital of 13.25%.

31.2.2 Step 2: Clean-up the financial statements

Because of various issues mentioned in the question, we need to clean-up the financial statements to come up with a truer picture of the actual performance of the business:

Particulars	Last year	Adjusted	Remarks
Revenues	1,200	1,200	
- Operating expenses	120	-	Leases are financial expenses
- Wages	200	350	Hire a chef for \$150K/ year
- Material	300	300	
- Other operating expense	180	180	
Operating income	400	370	
- Interest expense	-	70	7.5% of 928 (see below)
Taxable income	400	300	
- Taxes	160	120	
Net income	240	180	
Debt	-	928	PV of \$120 for 12 years @ 7.5%

31.2.3 Assess the impact of the “key” person

- Part of the draw of the restaurant comes from the current chef. It is possible (and probable) that if he sells and moves on, there will be a drop off in revenues. If you are buying the restaurant, you should consider this drop off when valuing the restaurant.
- Thus, if 20% of the customers are drawn to the restaurant because of the chef’s reputation, the expected operating income will be lower if the chef leaves.

- Adjusted operating income (existing chef) = \$370,000
- Operating income (adjusted for chef departure) = \$296,000 (less by 20%)
- To mitigate this risk, you might want the existing chef to stay on for a transition period (say 2 years) till the time the new management/ chef build credibility amongst their customers.

31.2.4 Step 4: Don't forget valuation fundamentals

- To complete the valuation, you need to make assumptions about expected growth rate. This assumption has to be consistent with reinvestment assumptions:

$$\text{Reinvestment rate} = \frac{\text{Growth}}{\text{Return on capital}}$$

- 2% perpetual growth rate and a return on capital of 20% yields a reinvestment rate of 10%.
- Even if the restaurant does not grow in size, this reinvestment is what you need to make to keep the restaurant both:
 - Looking good (remodelling); and
 - Working well (new ovens and appliances)

31.2.5 Step 5: Complete the valuation

Inputs in the valuation:

- Adjusted EBIT = \$296,000
- Tax rate = 40%
- Cost of capital = 13.25%
- Expected growth rate = 2%
- Reinvestment rate = 10%

Particulars	Working	Value
FCFF	$FCFF = 296 * (1 + 2\%) * (1 - 40\%) * (1 - 10\%)$	163.04
Value of restaurant	$V = \frac{163.04}{(13.25\% - 2\%)}$	1,449.22
- Value of leases		928.23
Value of equity		520.98

31.2.6 Step 6: Consider the effect of ILLIQUIDITY²²

The reason you should worry about illiquidity is buyer's remorse. In general, illiquidity is treated as a constant discount from value. However, it should vary across:

Companies	Time	Buyers
Money making/ cash flow generating businesses should worry less about liquidity than unprofitable/ negative cash flow companies.	Liquidity matters more at times of crisis and less in normal times. When everyone wants cash, liquidity matters a lot.	Long term buyers with less cash needs should care less about liquidity than short term buyers with greater cash needs.

Based on the regression equation discussed in Appendix A, we get an illiquidity discount of 12.88% for the restaurant.

Therefore, value of equity in the restaurant = $520.98 \times (1 - 0.1288) = \$454K$

²² See Appendix B for a detailed discussion on estimating an illiquidity discount

32 Private company sold to a public company

As the owner of the restaurant in the previous example, you would be really disappointed if someone offered you a value of just \$454,000 for your business which earned a net income of \$240,000 last year.

So what can you do – find a different buyer because it is the discounts and non-diversification of the investment banker that is driving the value of your restaurant so low.

32.1 Estimating the discount rate

Since a public buyer does not have to worry about firm-specific risk, we can re-compute the cost of capital for a public buyer as follows:

Particulars	Private buyer	Public buyer
Unlevered beta	2.36	1.18
Debt-to-equity ratio	14.33%	14.33%
Tax rate	40%	40%
Pre-tax cost of debt	7.50%	7.50%
Levered beta	2.56	1.28
Risk free rate	4.25%	4.25%
Equity risk premium	4%	4%
Cost of equity	14.50%	9.38%
After-tax cost of debt	4.50%	4.50%
Cost of capital	13.25%	8.76%

32.2 Revaluing the restaurant to the public buyer

A public buyer does not worry about illiquidity because investors in the public company can readily sell the shares of the public company on a stock exchange if they wish to.

Particulars	Private buyer	Public buyer
Adjusted EBIT	370	370
Key person discount	20%	20%
EBIT	296	296

Expected growth rate	2%	2%
Return on capital	20%	20%
Reinvestment rate	10%	10%
FCFF next year	163.04	163.04
Cost of capital	13.25%	8.76%
Value of the business	1,449.22	2,411.79
- Debt	928.23	928.23
Value of equity (before illiquidity)	520.99	1,483.56
- Illiquidity discount	12.88%	-
Value of equity	453.88	1,483.56

Lesson

If you are looking to sell your private business, look for a public buyer because you are likely to get a much higher value for your business than if you sell to another private buyer.

33 Private company for an IPO

Taking a private business public is similar to selling to a public buyer. However, there are specific issues in case of IPOs:

1. Proceeds that the firm gets from an IPO.
2. Claims from prior equity investors.
3. Investment banking guarantees.

33.1 Proceeds from an IPO

When the firm gets cash from an IPO, its implication in the valuation of the firm will depend on how the cash is used.

33.1.1 Deployment of proceeds

The proceeds from an IPO can be:

- Taken out of the firm by the existing owners;
- Used to pay down debt and other obligations; or
- Held as cash by the company to cover future reinvestment needs.

33.1.2 Dealing with proceeds in a valuation

How you deal with the issuance will depend upon how the proceeds are used mentioned above:

Taken out of the firm	Used to pay down debt	Held as cash for future reinvestment
Ignore	Change debt ratio which may change the cost of capital and value of the firm	Add the cash proceeds to the DCF valuation of the company

33.2 Claims of prior equity investors

When a private firm goes public, there are already equity investors in the firm, including the founders, VCs, etc. In some cases, these investors can have warrants, options or other special claims on the equity of the firm.

If existing equity investors have special claims on the equity, the value of equity per share has to be affected by these claims. Specifically, these options need to be:

- Valued at the time of offering; and
- Value of equity reduced by the option value before determining value per share.

33.3 The investment banking guarantee

Almost all IPOs are managed by investment banks and are backed by a pricing guarantee, where the investment banker guarantees the offering price to the issuer.

If the price at which the issuance is made is lower than the guaranteed price, the investment banker will buy shares at the guaranteed price and potentially bear the loss.

34 Private to VC to Public

34.1 Propositions

Proposition 1: The value of a private business that is expected to transition to a publicly traded company will be higher than the value of an otherwise similar private business that does not expect to make this transition.

- Private businesses in sectors that are “hot” in terms of going public (think social media in 2014) will be worth more than private businesses in less sexy areas;
- As IPOs boom (bust) private company valuations will increase (decrease);
- Private companies in countries that have easy access to public markets will have higher value than companies in countries without that access.

Proposition 2: The value of a private business that expects to make the transition to a public company sooner will be higher than the value of an otherwise similar company that will take longer.

- Private businesses will be worth more if companies are able to go public earlier in their life cycle.

35 Closing thoughts on private company valuation

- The value of a private company will depend on the potential buyer.
- If you are the seller of a private business, you will maximize value if you can sell to:
 - A long term investor;
 - Who is well diversified (or whose investors are); and
 - Does not think too highly of you (as a person).
- If you are valuing a private business for legal purposes (tax or divorce court), the assumptions you use and the value you arrive at will depend on which side of the legal divide you are on.
- As a final proposition, always keep in mind that the owner of a private business has the option of investing his wealth in publicly traded stocks. There has to be a relationship

between what you can earn on those investments and what you demand as a return on your business.

Appendix A: Primer on option pricing models

The driving force in all of option pricing is the notion of a REPLICATING PORTFOLIO.

Prior to 1971

- **Value the option like any other asset:** expected cash flows, project out the cash flows, adjust for probabilities and take the present value of the expected cash flows to come up with a value of the option.
- Therefore, options were valued like a normal stock or a bond.

Post 1971 (Fischer Black and Myron Scholes)

- **The insight:** We can create something that looks exactly like an option in terms of cash flows by using the underlying asset and either borrowing or lending what is called the replicating portfolio.
- **The principle of arbitrage:** If you have two investments with the same cash flows, they should trade at the SAME PRICE. In every option pricing model, you try to:
 - create a replicating portfolio;
 - see how much it will cost you to create that replicating portfolio; and
 - The option has to trade at that same value because if it did not, you have a riskless way of making money.

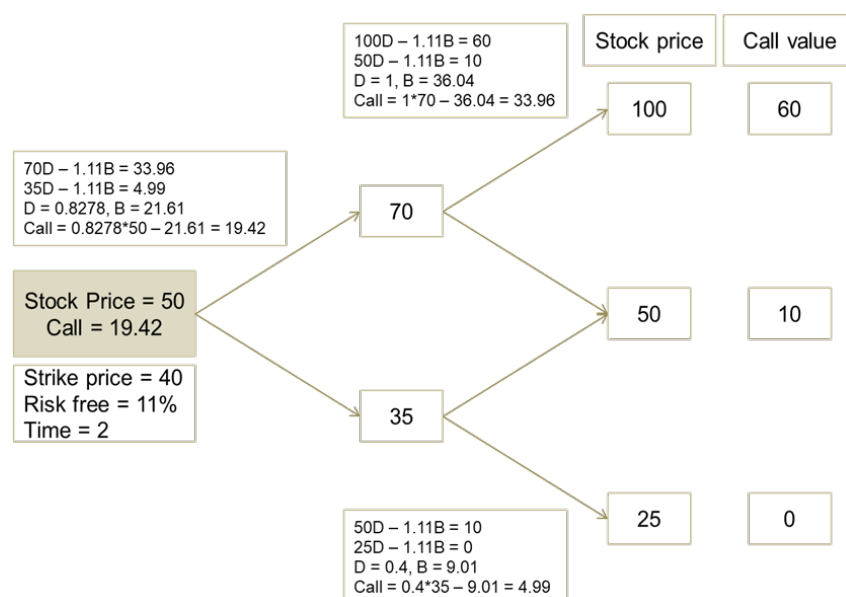
Therefore, every option pricing model is a search for that replicating portfolio.

Creating a replicating portfolio

The objective in creating a replicating portfolio is to use a combination of risk-free borrowing/lending and the underlying asset to create the same cash flows as the option being valued:

- Call option = Borrowing + Buying D of underlying stock
- Put option = Lending + Selling short D on underlying asset

The Binomial model



The limiting distribution

As time interval is shortened, the limiting distribution, as $t \rightarrow 0$, can take one of two forms:

T ~ 0	
Price changes become smaller	Price changes remain large
Normal distribution and the price process is a continuous one	Poisson distribution, i.e. a distribution that allows for price jumps

The Black-Scholes model is not a different model. It essentially is a binomial model which applies when the limiting distribution is the normal distribution (as time goes to 0), and explicitly assumes that the price process is continuous and that there are no jumps in asset prices. Therefore, a Black-Scholes option pricing model suffers from the following:

Deep out-of-the money	Underestimate	Because the stock will take much more time than normal to move significantly in either direction (up or down).
Deep in-the-money	Overestimate	

The Black Scholes Formula

$$C = S * e^{-yt} * N(d_1) - K * e^{-rt} * N(d_2)$$

$$d_1 = \frac{\left[\ln\left(\frac{S}{K}\right) + \left(r - y + \frac{\sigma^2}{2}\right) * t \right]}{\sigma * \sqrt{t}}$$

$$d_2 = d_1 - \sigma\sqrt{t}$$

Appendix B: Estimating an illiquidity discount

Standard approach: Illiquidity discount based on illiquid publicly traded assets

Restricted stock: These are stock issued by publicly traded companies to the market that bypass the SEC registration process but the stock cannot be traded for one year after the issue.

$$Discount_{restricted} = Stock\ price - Price\ on\ restricted\ stock$$

Most studies give an average/ median discount of somewhere around 35% which is used as an illiquidity discount in coming up with the value of private companies.

Pre-IPO transactions: These are transactions prior to initial public offerings where equity investors in the private firm buy (sell) each other's stakes

$$Discount_{IPO} = Stock\ price - Price\ on\ pre\ IPO\ transaction$$

The “sampling” problem

However, both the above studies suffer from a significant sampling bias:

- The companies that make restricted stock offerings are likely to be small, troubled firms that have run out of conventional financing options.
- The types of IPOs where equity investors sell their stake in the five months prior to the IPO at a huge discount are likely to be IPOs that have significant pricing uncertainty associated with them.

In fact, the magnitude of the sampling bias was estimated by comparing the discount on all private placements to the discount on restricted stock offerings. One study concluded that the “illiquidity” alone accounted for less than 10%, leaving the remaining 20% - 25% to be explained by sampling problems.

The solution: Using the bid-ask spread as a measure of illiquidity

- We can regress the bid-ask spread (as a % of price) against variables that can be measured for a private firm which are also available for publicly traded firms (such as revenues, cash flow generating capacity, type of assets, variance in operating income, etc.).
- For instance, for data since 2000, we get the following regression:

$$Spread = 0.145 - 0.0022 * \ln(Annual\ Revenues) - 0.015 * (DERN) - 0.016 * \left(\frac{Cash}{Firm\ Value} \right) - 0.11 * \left(\frac{\$ \text{ monthly trading volume}}{Firm\ Value} \right)$$

Where, DERN = dummy variable for positive earnings, i.e. DERN = 0 if negative and 1 if positive

For a private firm, the monthly traded volume can be considered as 0.